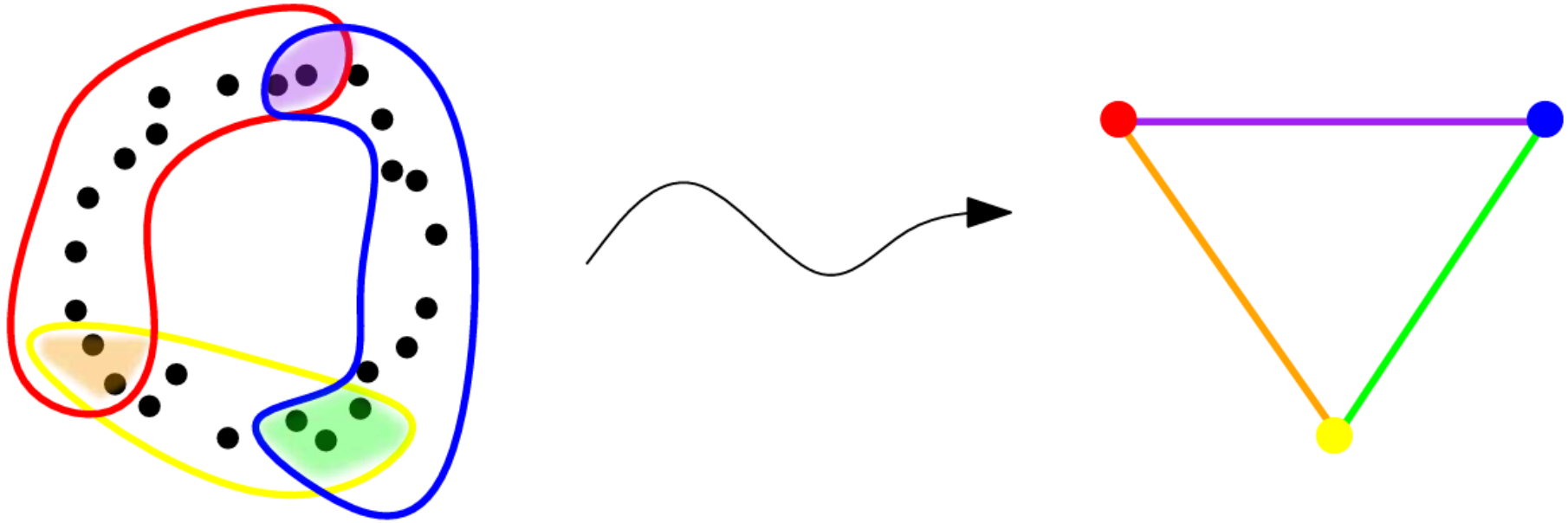
Mapper Algorithm



Idea:

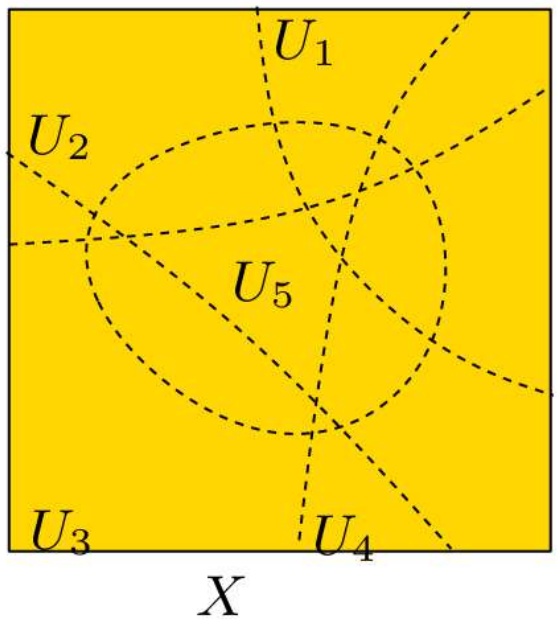
- Group data points in "local clusters"
- Summarize the data through the combinatorial/topological structure of intersection patterns of "clusters"

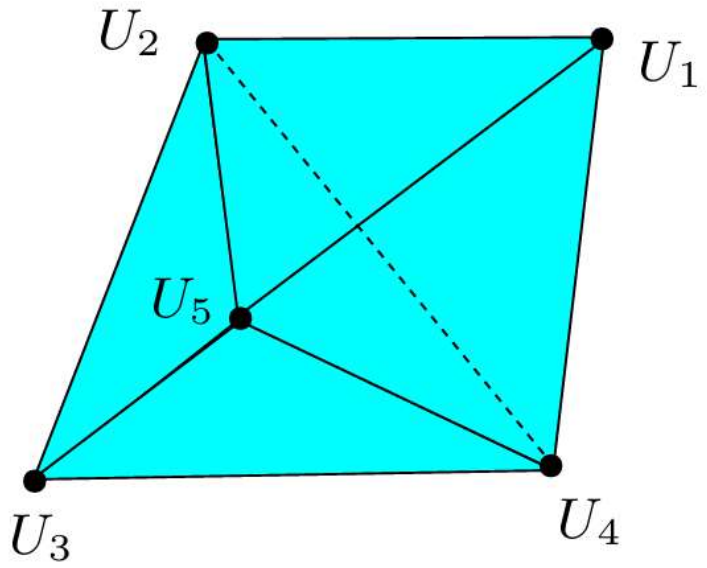
Goal: Do it in a way that preserves (some of) the topological features of the data.

source: Frederic Chazal NerveUnionofBalls.pdf

Given a cover of a topological space X, $\mathcal{U} = (U_i)_{i \in I}$, its nerve is the abstract simplicial complex $C(\mathcal{U})$ whose vertex set is $\mathcal{U}$ and such that

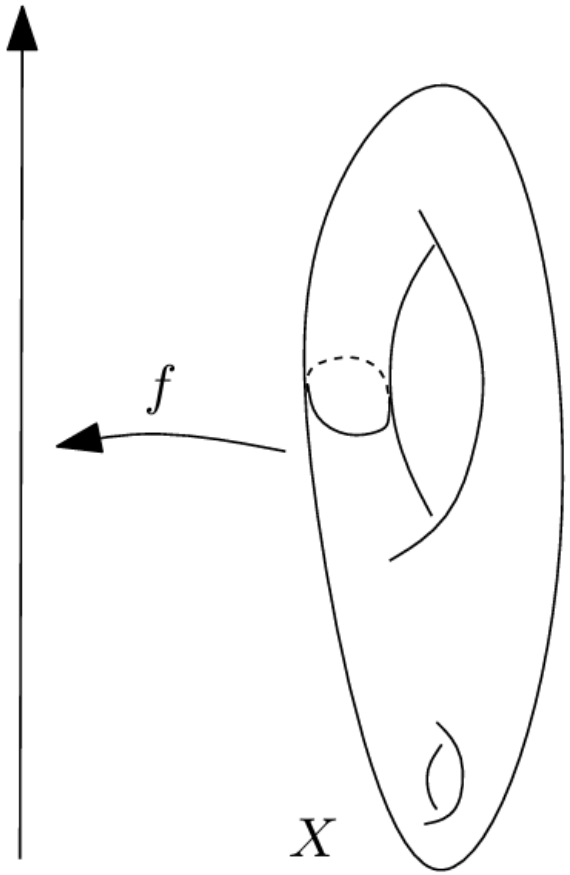
$$\sigma = [U_{i_0}, U_{i_1}, \cdots, U_{i_k}] \in C(\mathcal{U}) \text{ if and only if } \cap_{j=0}^{k} U_{i_j} \neq \emptyset.$$



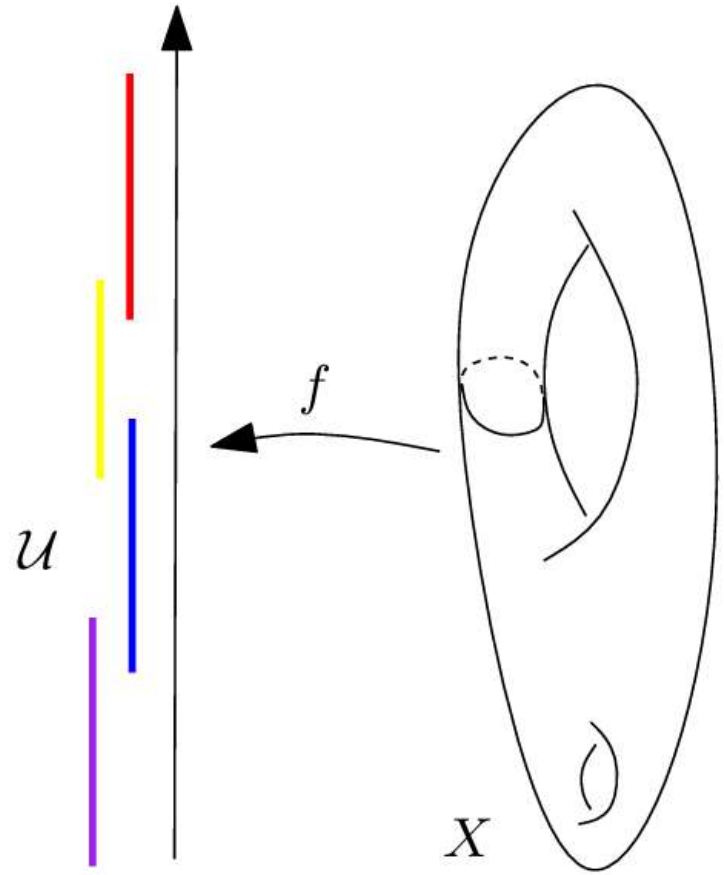


The Nerve Theorem:
Let $\mathcal{U} = (U_i)_{i \in I}$ be a finite open cover of a subset X of $\mathbb{R}^d$ such that any intersection of the U_i's is either empty or contractible. Then X and $C(\mathcal{U})$ are homotopy equivalent.

source: Frederic Chazal NerveUnionofBalls.pdf

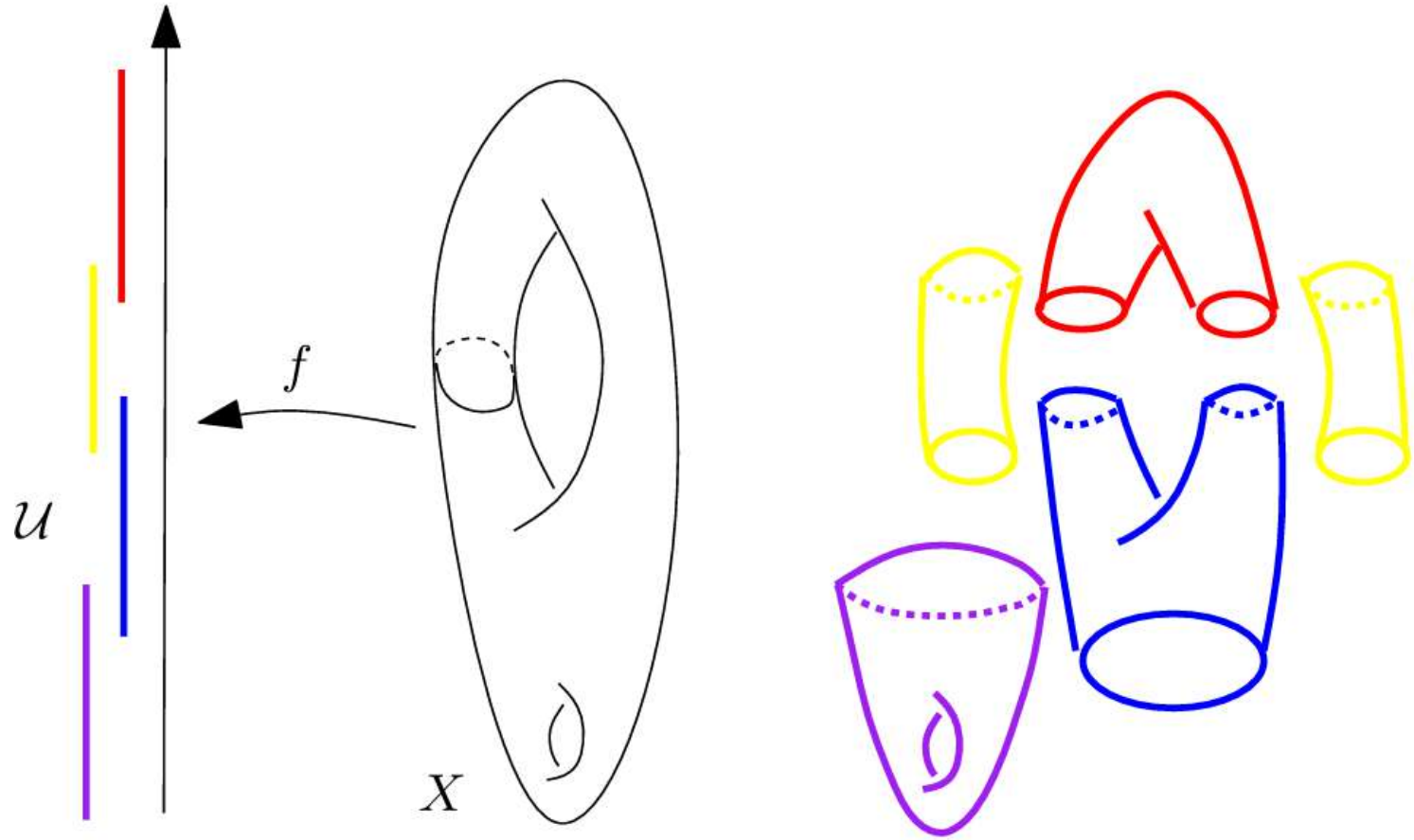


source: Frederic Chazal NerveUnionofBalls.pdf



Let $f : X \to \mathbb{R}$ (or $\mathbb{R}^d$) a continuous function where X is a topological space and let $\mathcal{U} = (U_i)_{i \in I}$ be a cover of $\mathbb{R}$ (or $\mathbb{R}^d$).

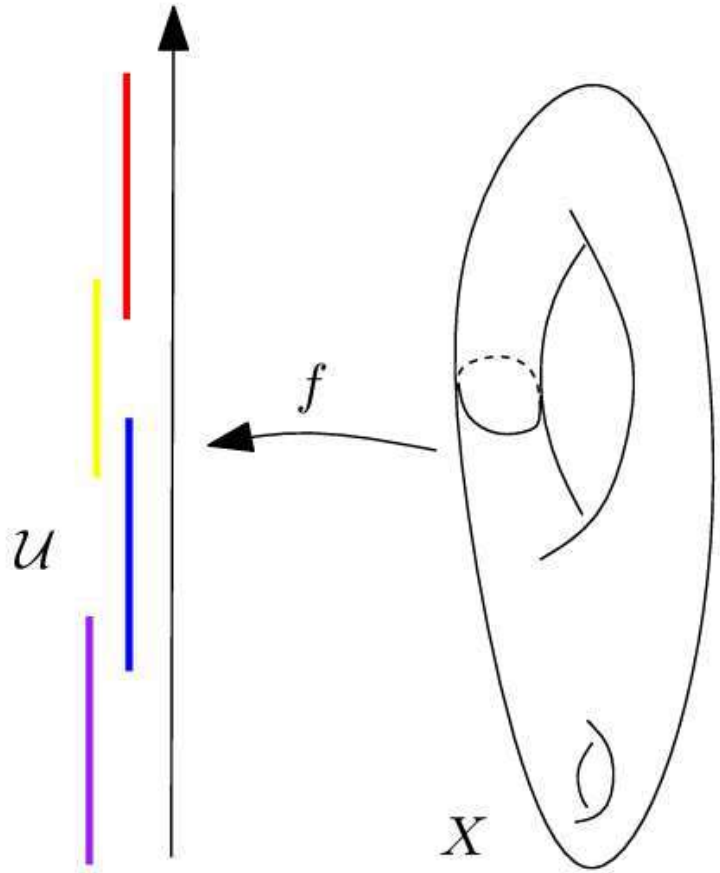
source: Frederic Chazal NerveUnionofBalls.pdf

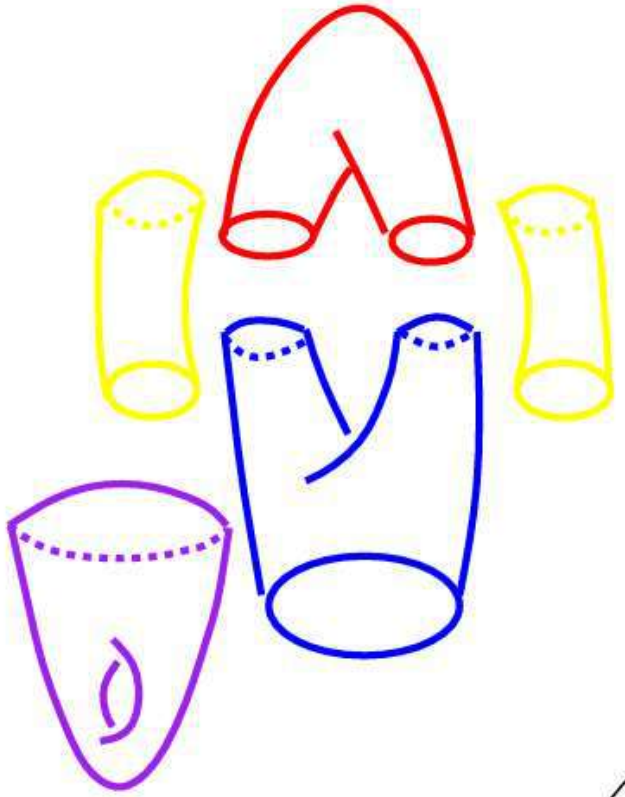


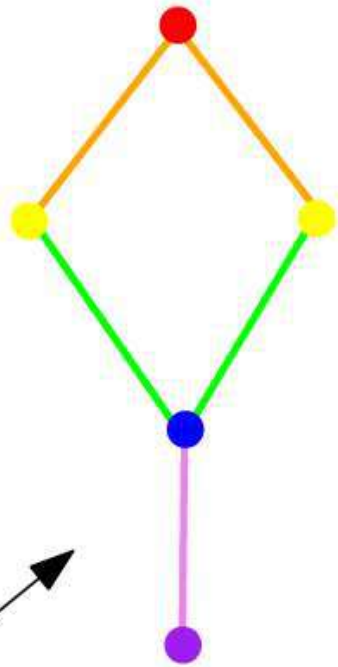
Let $f : X \to \mathbb{R}$ (or $\mathbb{R}^d$) a continuous function where X is a topological space and let $\mathcal{U} = (U_i)_{i \in I}$ be a cover of $\mathbb{R}$ (or $\mathbb{R}^d$).

The collection of open sets $(f^{-1}(U_i))_{i \in I}$ is the pull back cover of X induced by $(f, \mathcal{U})$.

source: Frederic Chazal NerveUnionofBalls.pdf







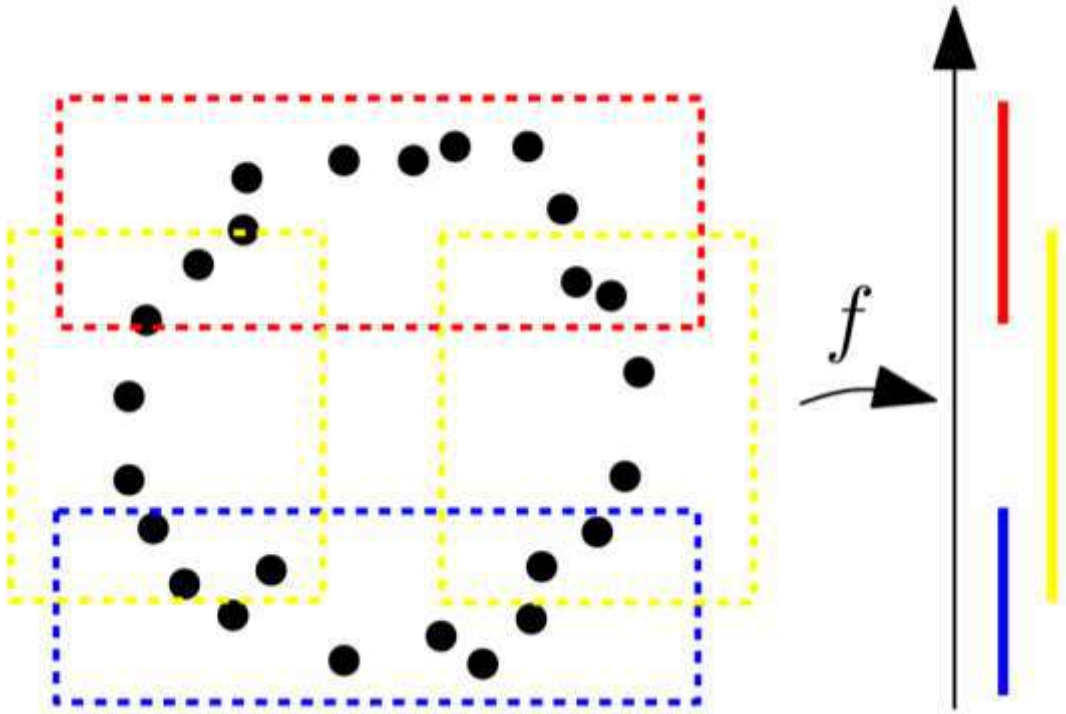
Take the connected components of the $f^{-1}(U_i)$, $i \in I \rightarrow$ the refined pull back cover.

Take the nerve of the refined cover.

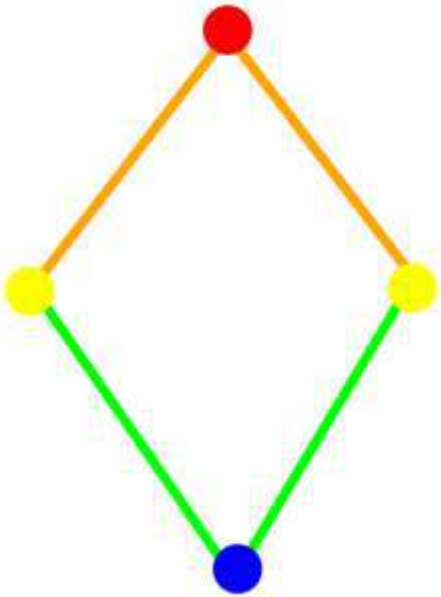
Warning: The nerve theorem does not apply in general!

source: Frederic Chazal NerveUnionofBalls.pdf

Mapper Algorithm



1. for each $U \in \mathcal{U}$, decompose $f^{-1}(U)$ into clusters $C_{U,1}, \cdots, C_{U,k_U}$.
2. Compute the nerve of the cover of X defined by the $C_{U,1}, \cdots, C_{U,k_U}$, $U \in \mathcal{U}$



Input:

- A data set X with a metric
- A function $f : X \to R$ or R^d
- A cover U of $f(X)$

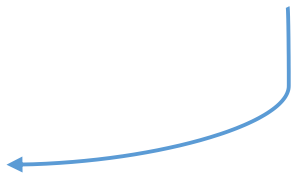
Output: a simplicial complex, the nerve

Often visualized as a graph for well-chosen covers:

- A vertex $v_{U,i}$ for each cluster $C_{U,i}$,
- An edge between $v_{U,i}$ and $v_{U',j}$ iff $C_{U,i} \cap C_{U',j} \neq \emptyset$.

source: Frederic Chazal NerveUnionofBalls.pdf

Mapper parameters: projection



Classical choices:

- density estimates
- centrality $f(x) = \sum_{y \in X} d(x, y)$
- excentricity $f(x) = \max_{y \in X} d(x, y)$
- PCA coordinates, NLDR coordinates,...
- Eigenfunctions of graph laplacians.
- data driven
- ...

Mapper parameters: projection, cover

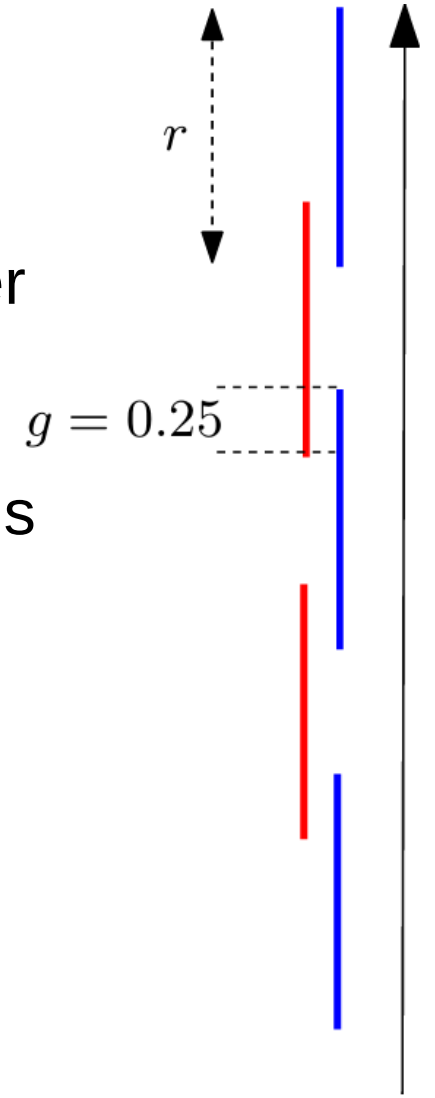
Classical choices:

- density estimates
- centrality $f(x) = \sum_{y \in X} d(x, y)$
- excentricity $f(x) = \max_{y \in X} d(x, y)$
- PCA coordinates, NLDR coordinates,...
- Eigenfunctions of graph laplacians.
- data driven
- ...

The resolution **r** = maximum diameter of an interval in U.

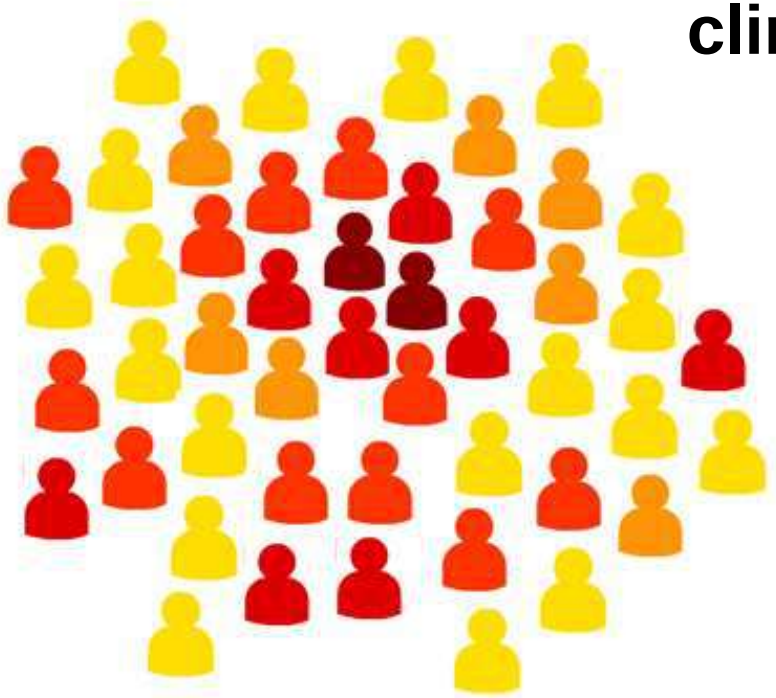
Number of intervals **N** in the cover.

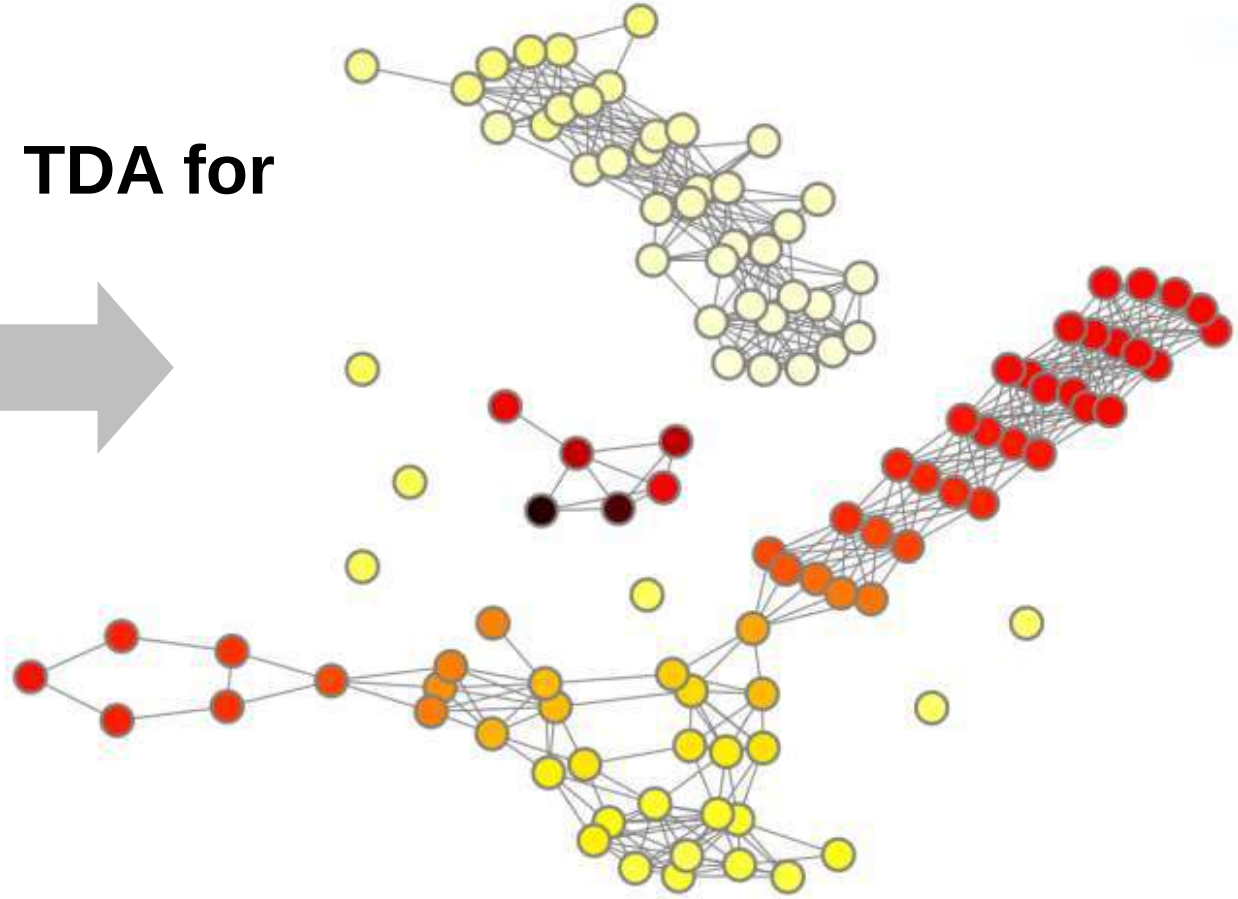
The gain **g** is the percentage of overlap between intervals (when they overlap)



Multiscale α-neighboring (MαN) graph approach

Graph-based TDA for clinical data

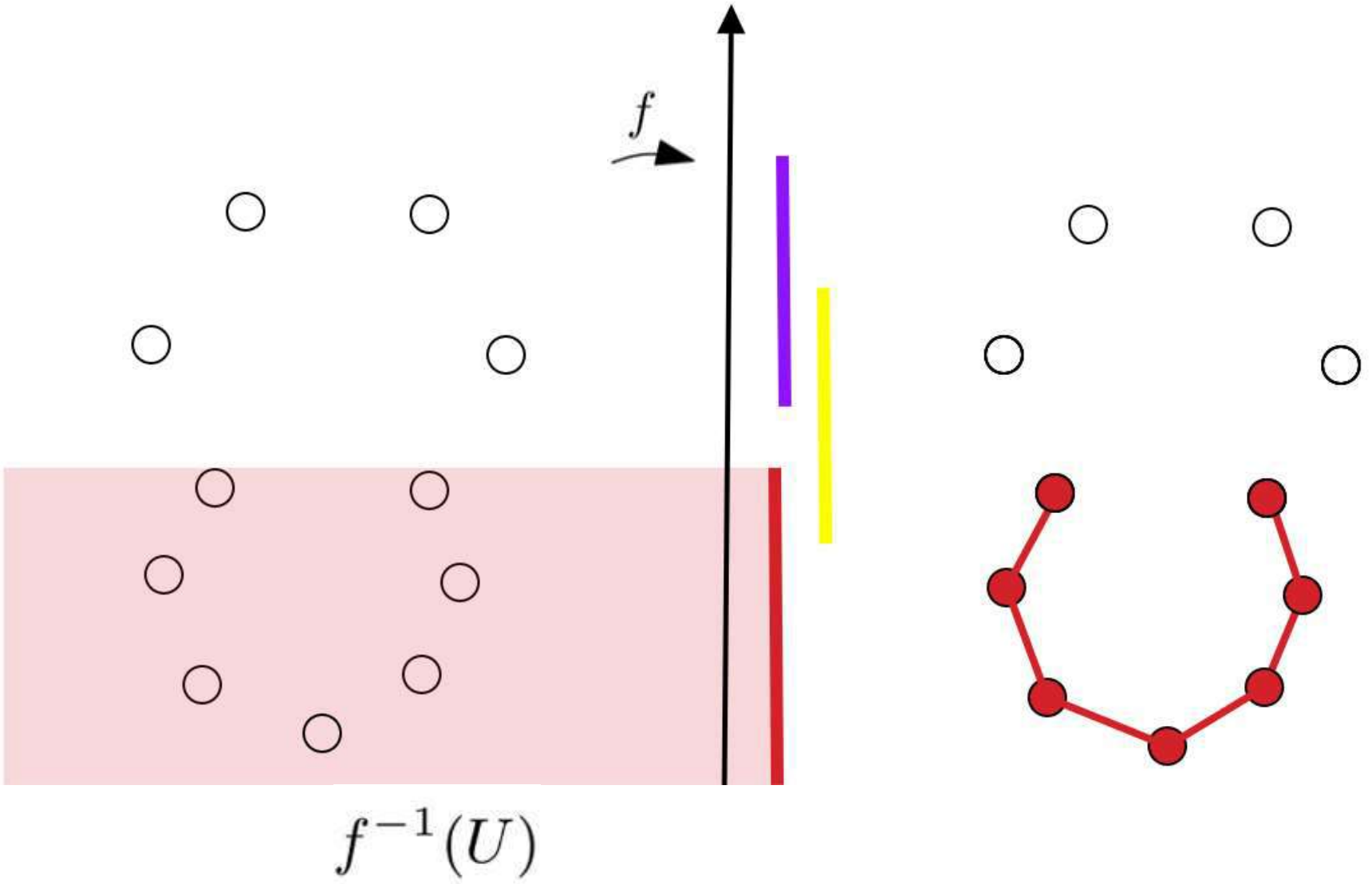




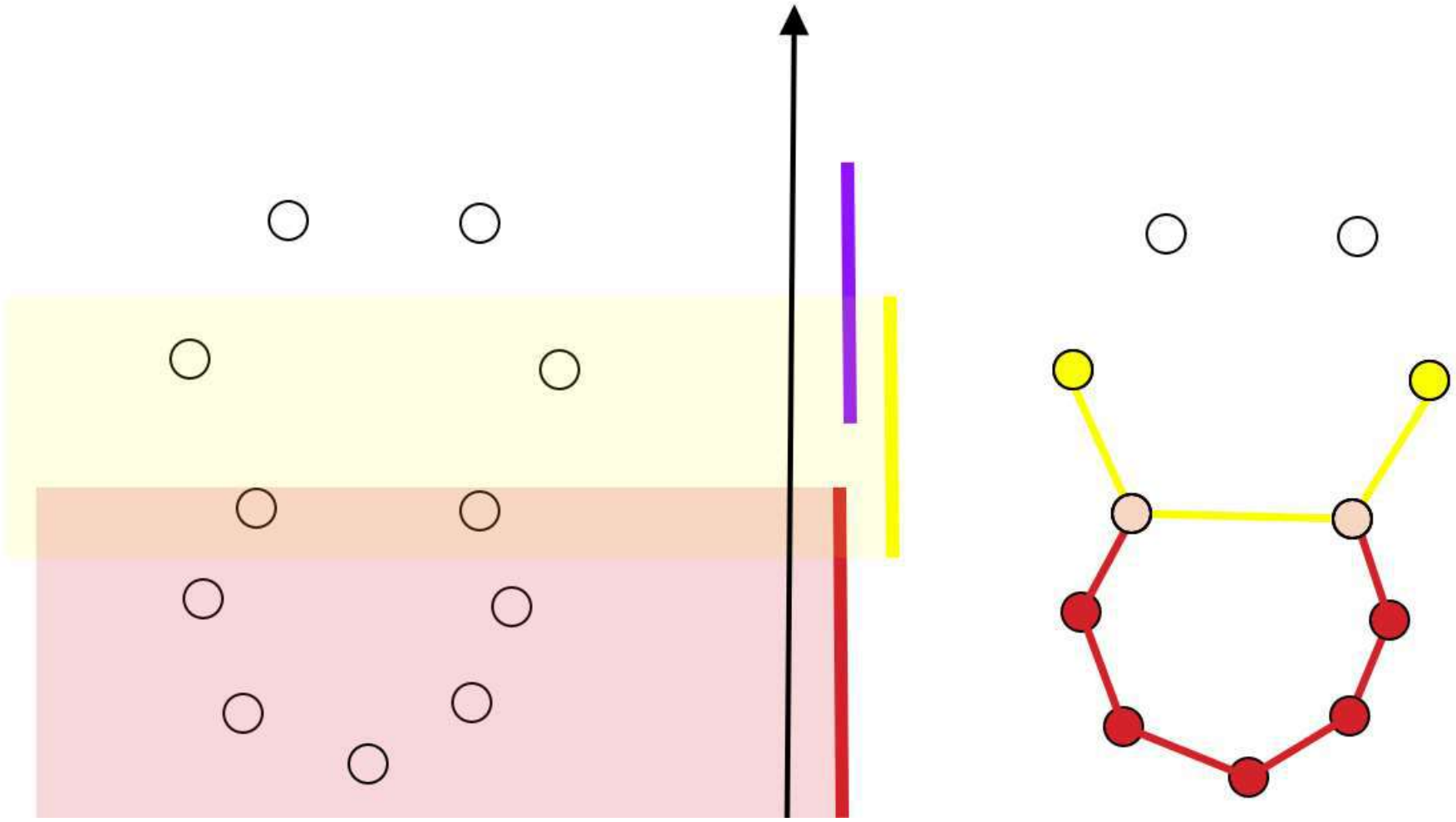
Each node represents one subject

Two nodes are connected if they are similar, within projection intervals, in terms of outcomes

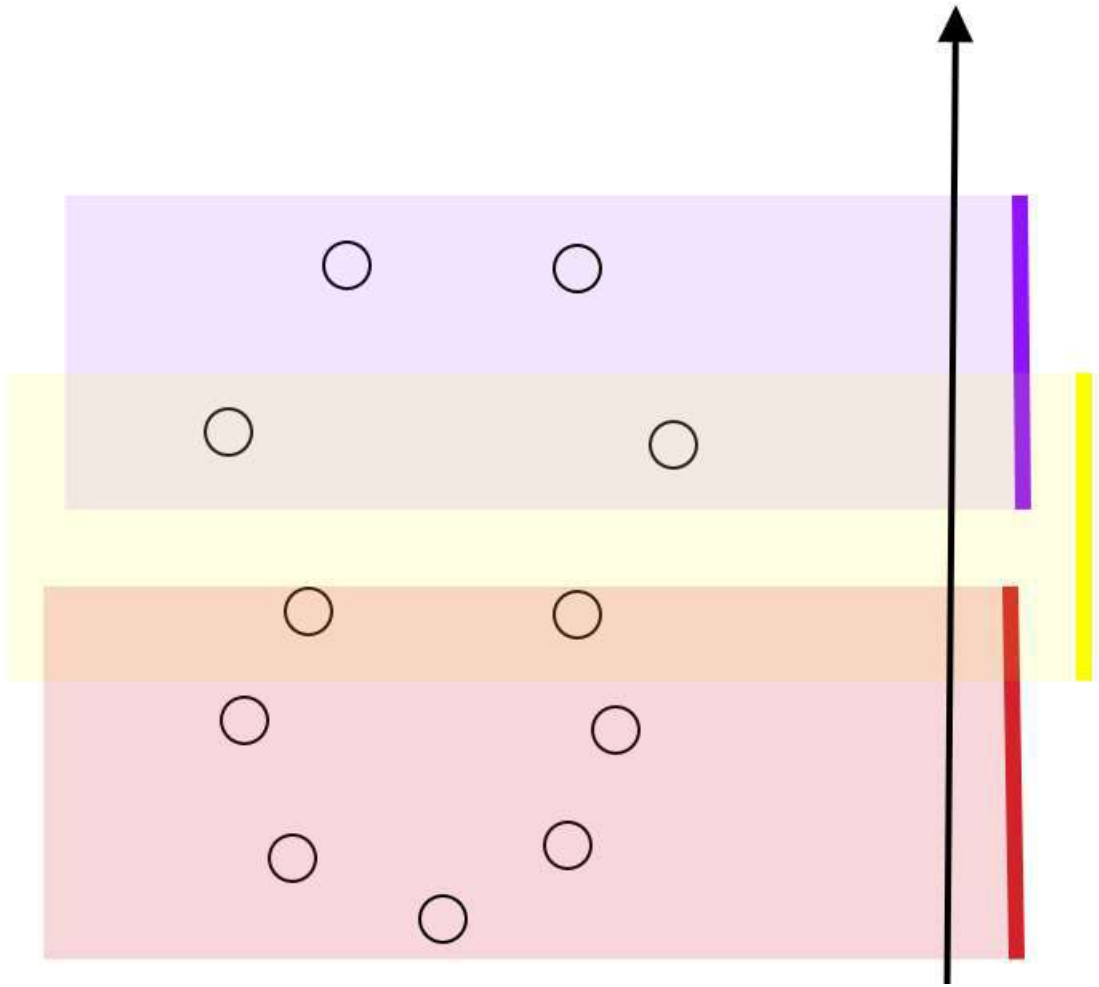
Building MαN graph

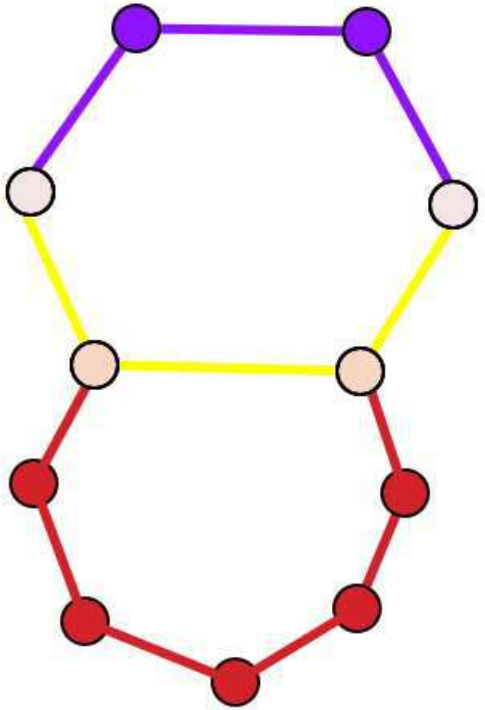


Building MαN graph



Building MαN graph

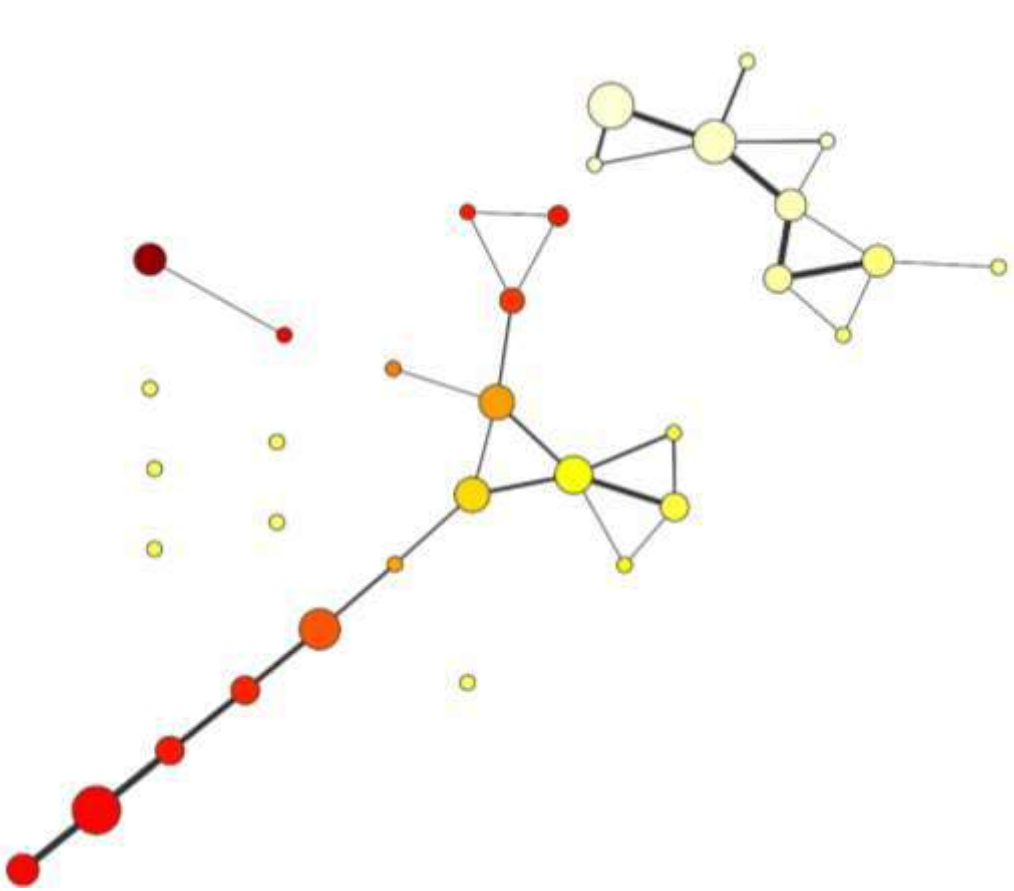


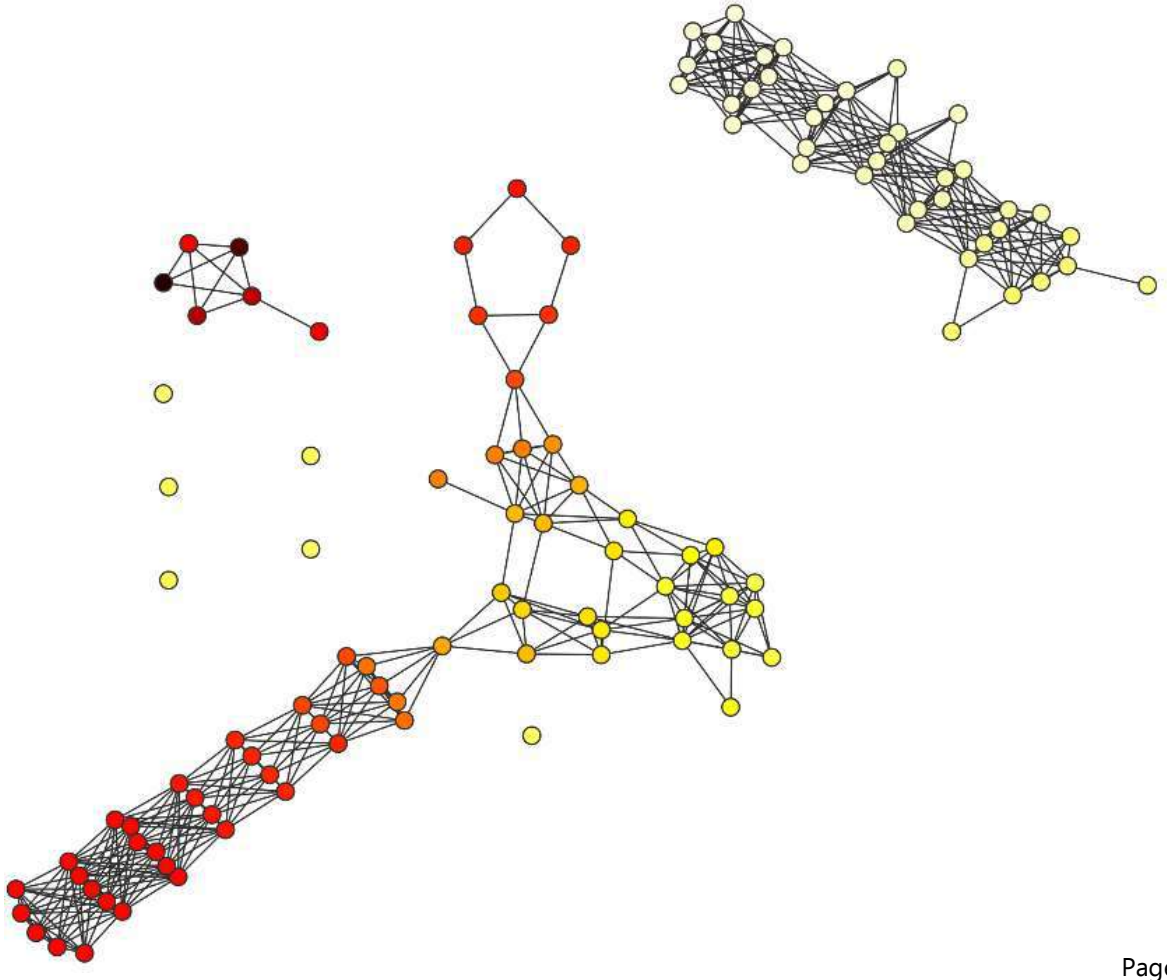


Mapper

MαN graph

Both use projection function and covering



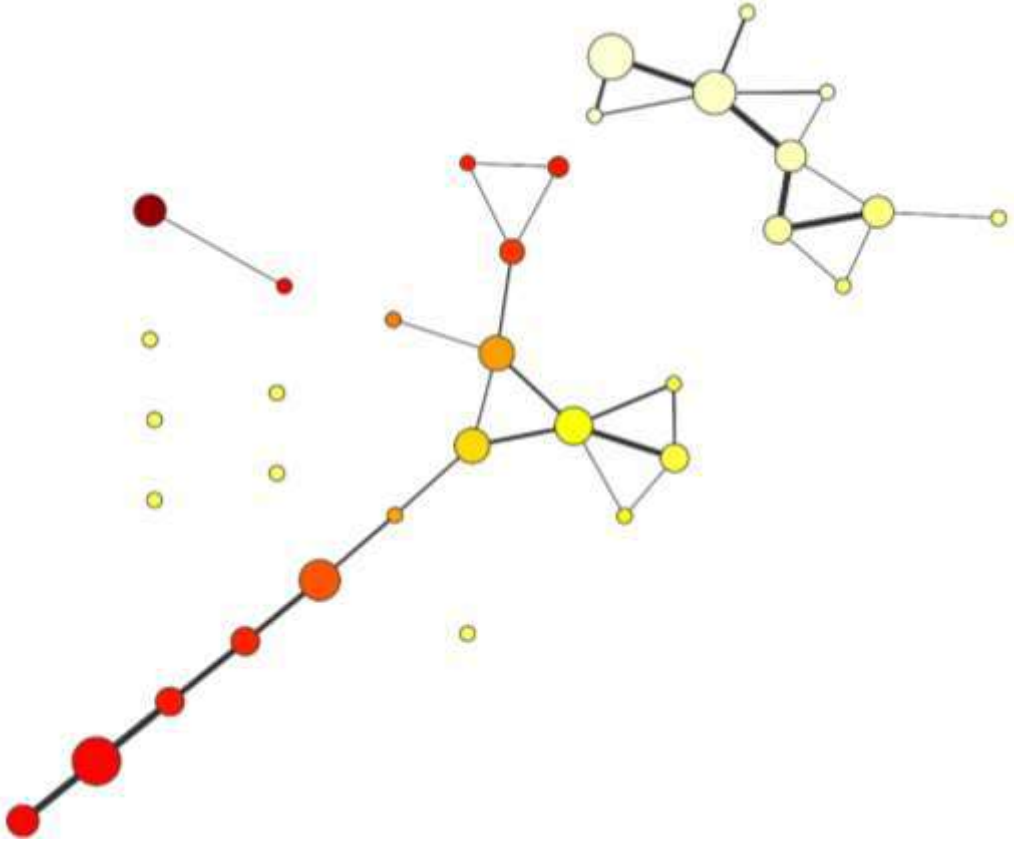


Mapper

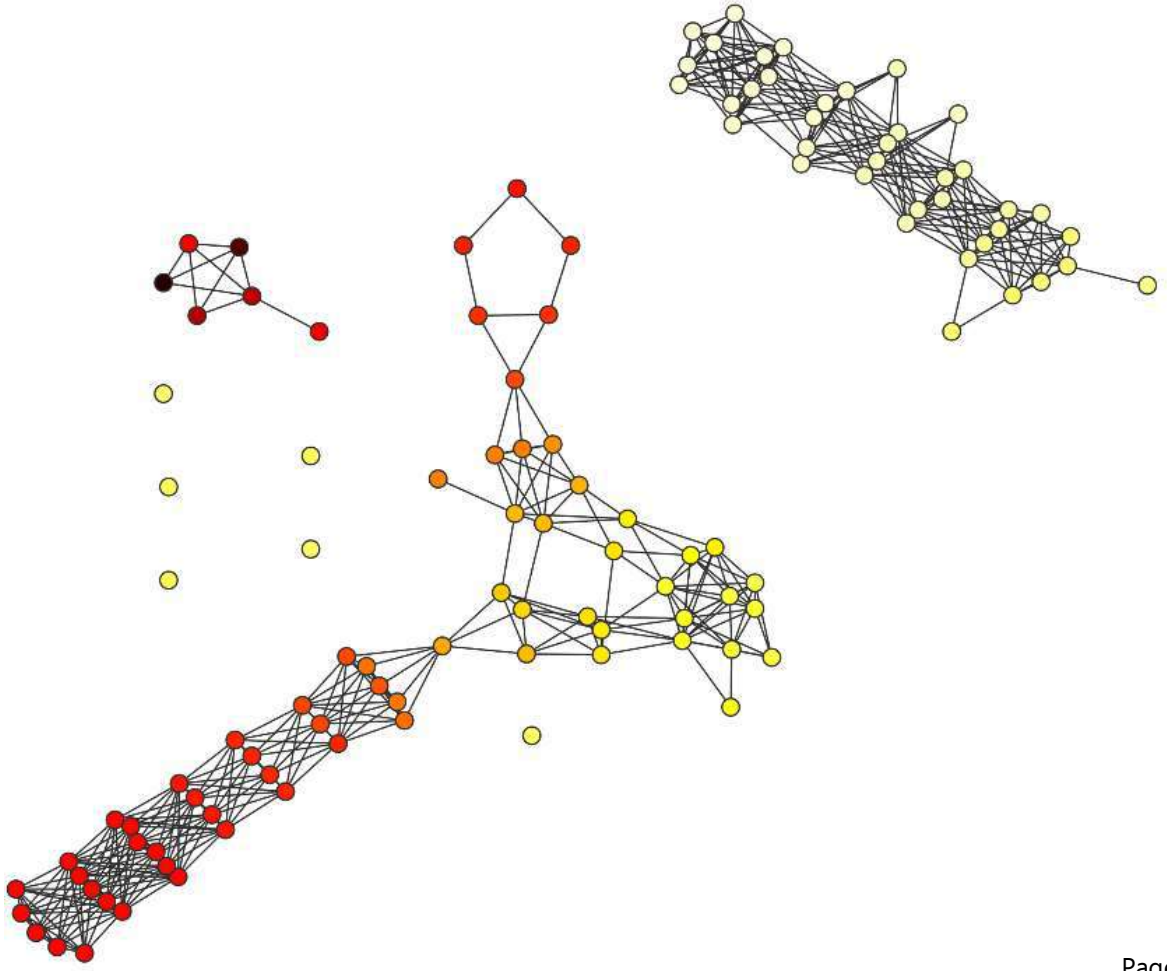
MαN graph

Both use projection function and covering

- Nodes correspond to clusters



- Nodes correspond to original data points



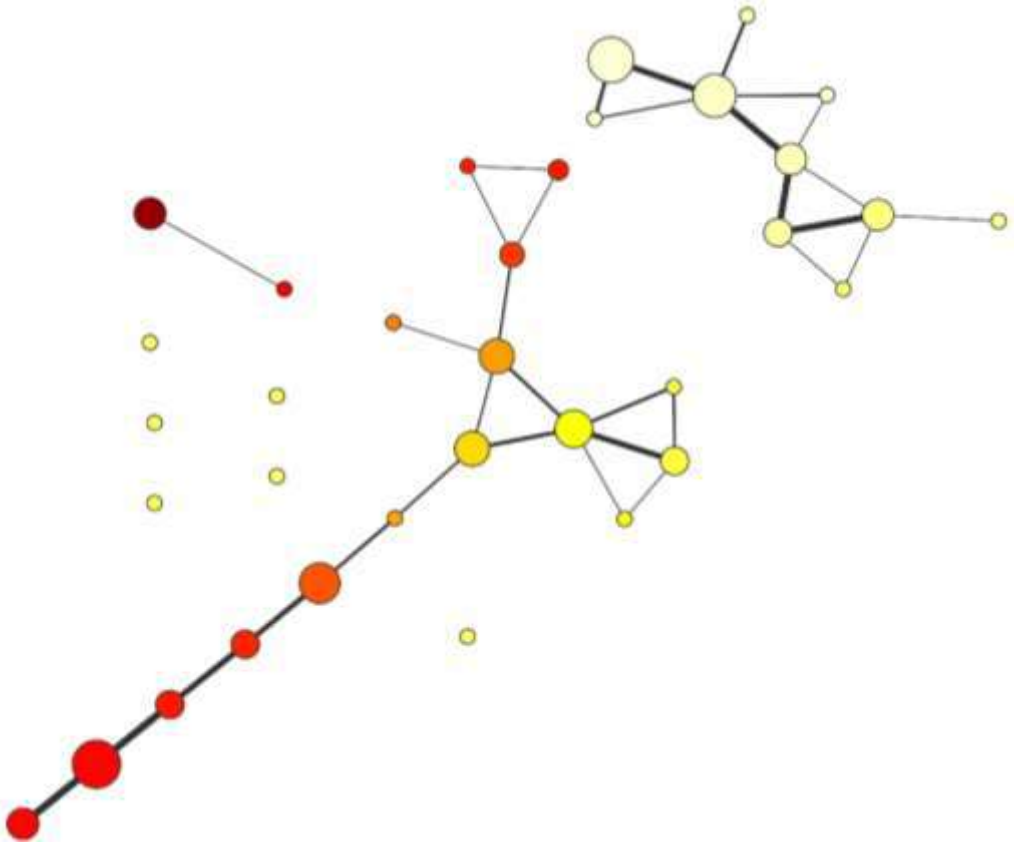
Mapper

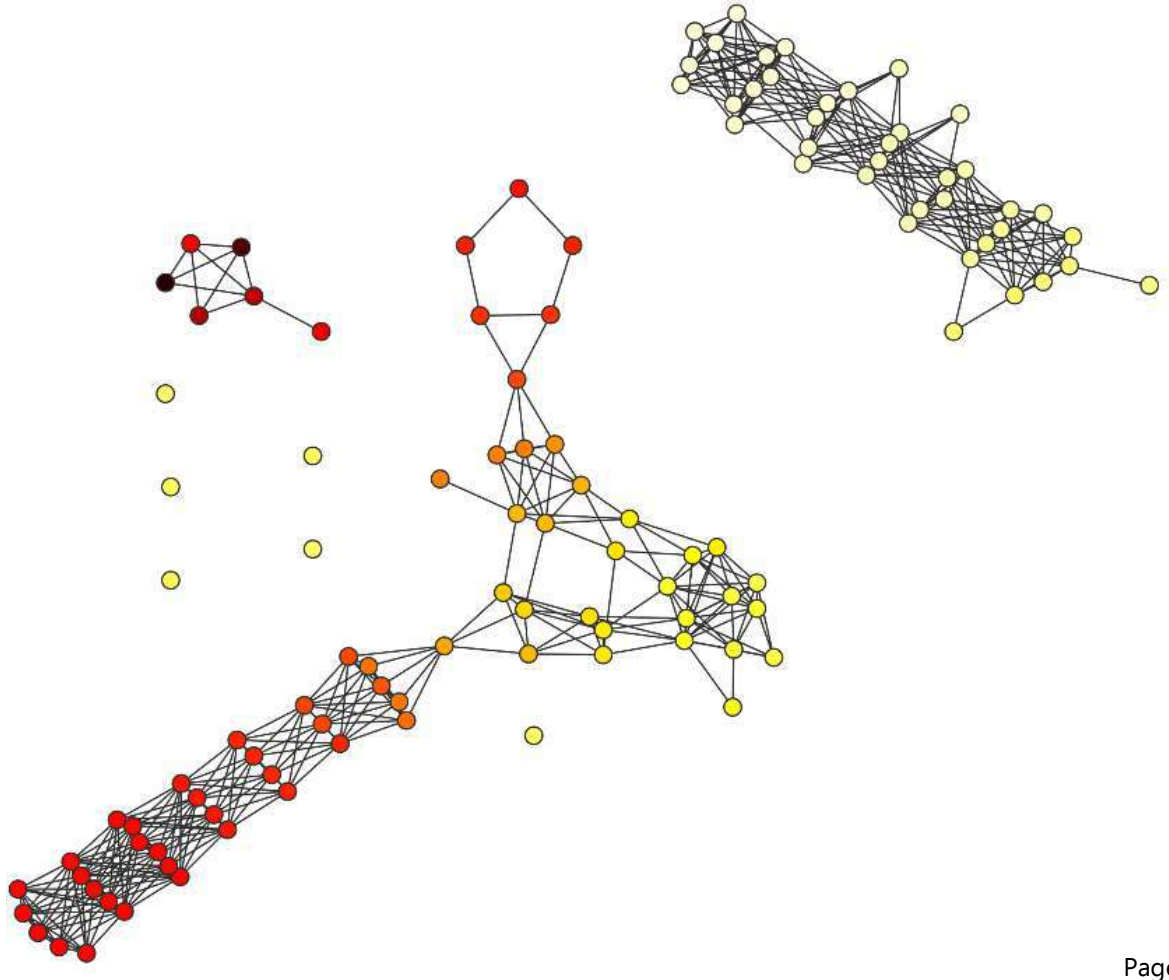
MαN graph

Both use projection function and covering

- Nodes are connected iff clusters intersect

- Nodes are connected iff similar *(by metric distance)*





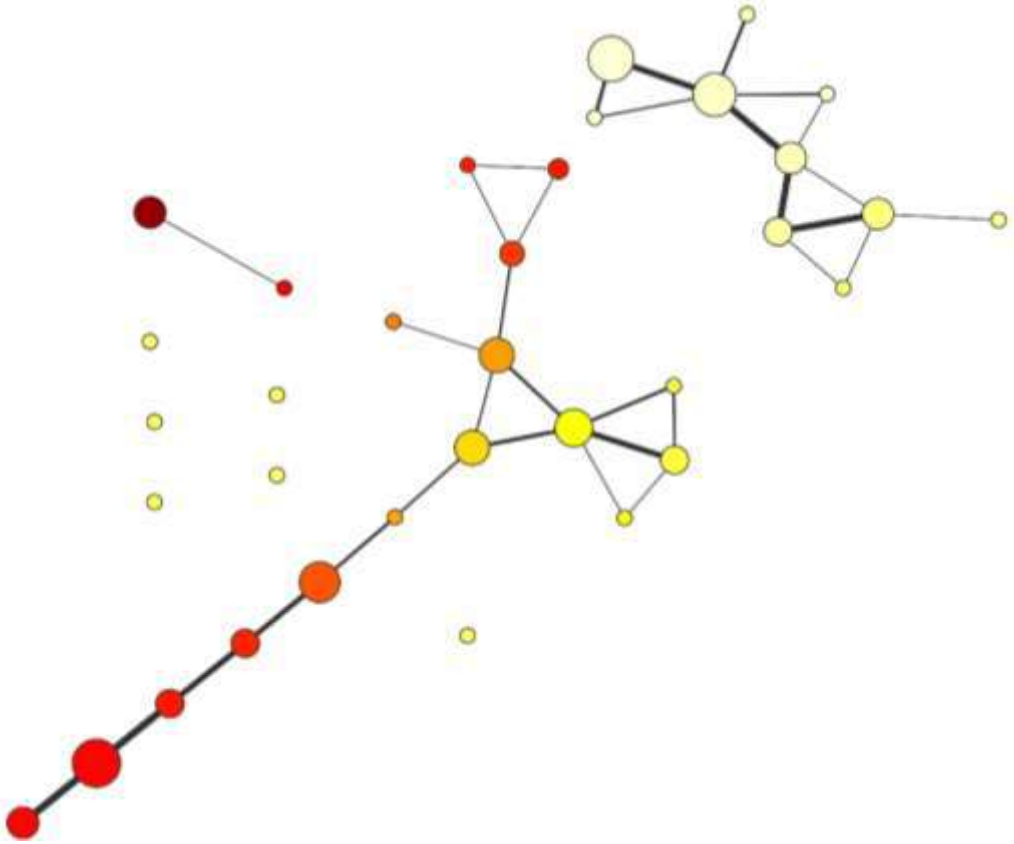
Mapper

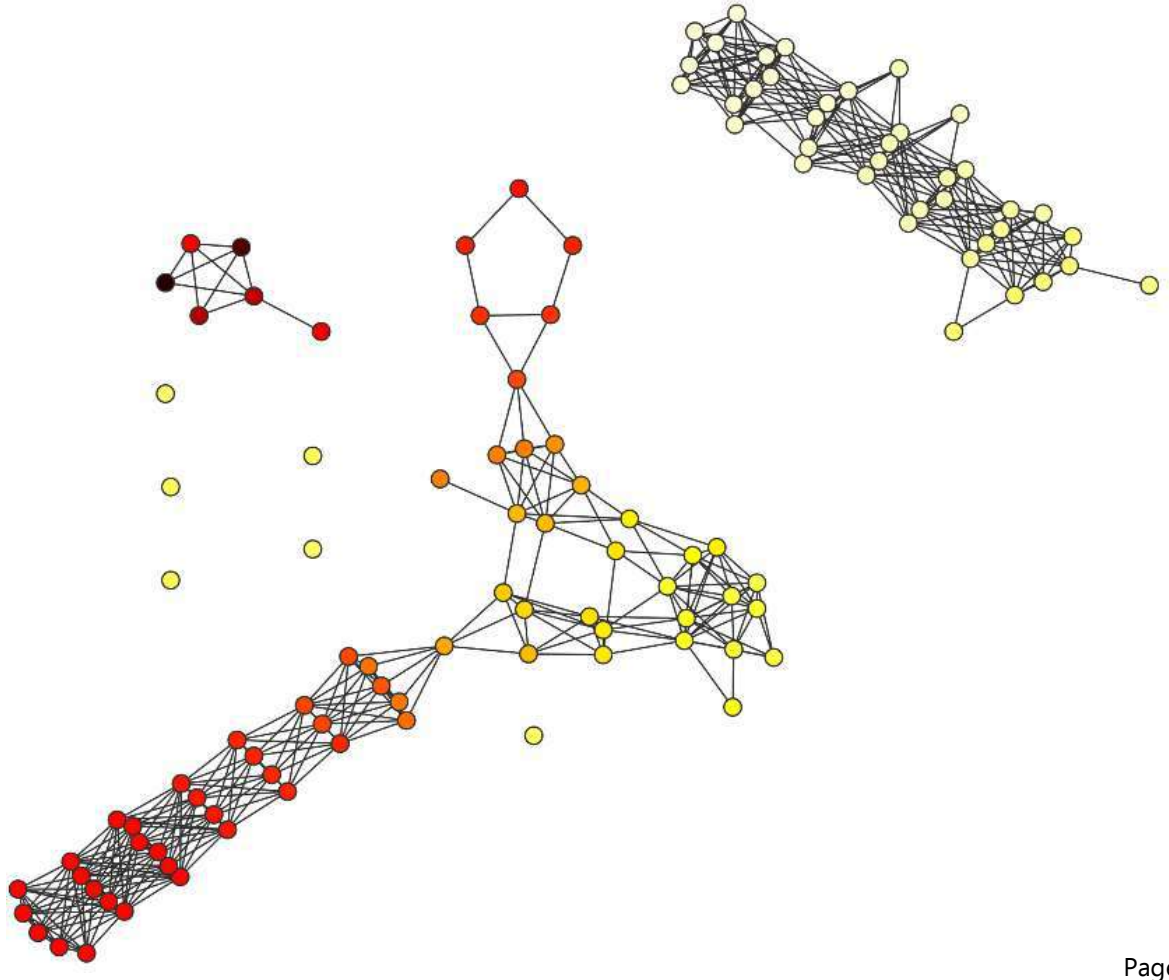
MαN graph

Both use projection function and covering

- Depends on clustering algorithms

- Depends on selection of scale thresholds



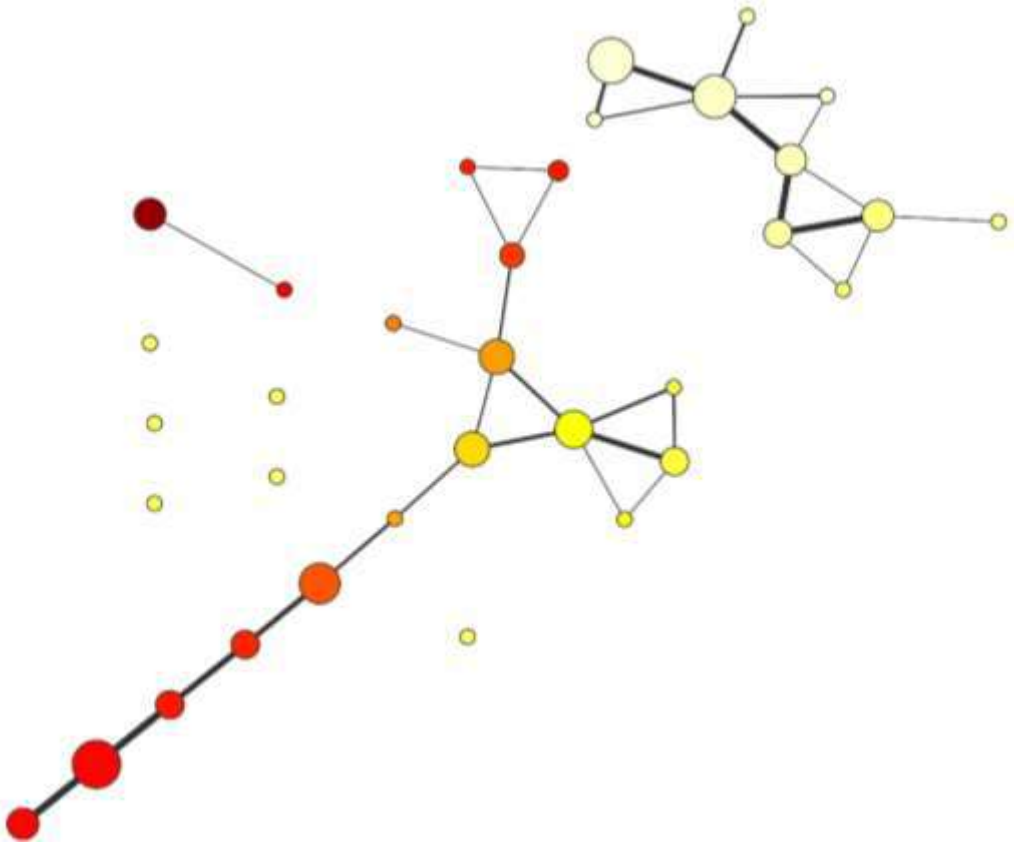


Mapper

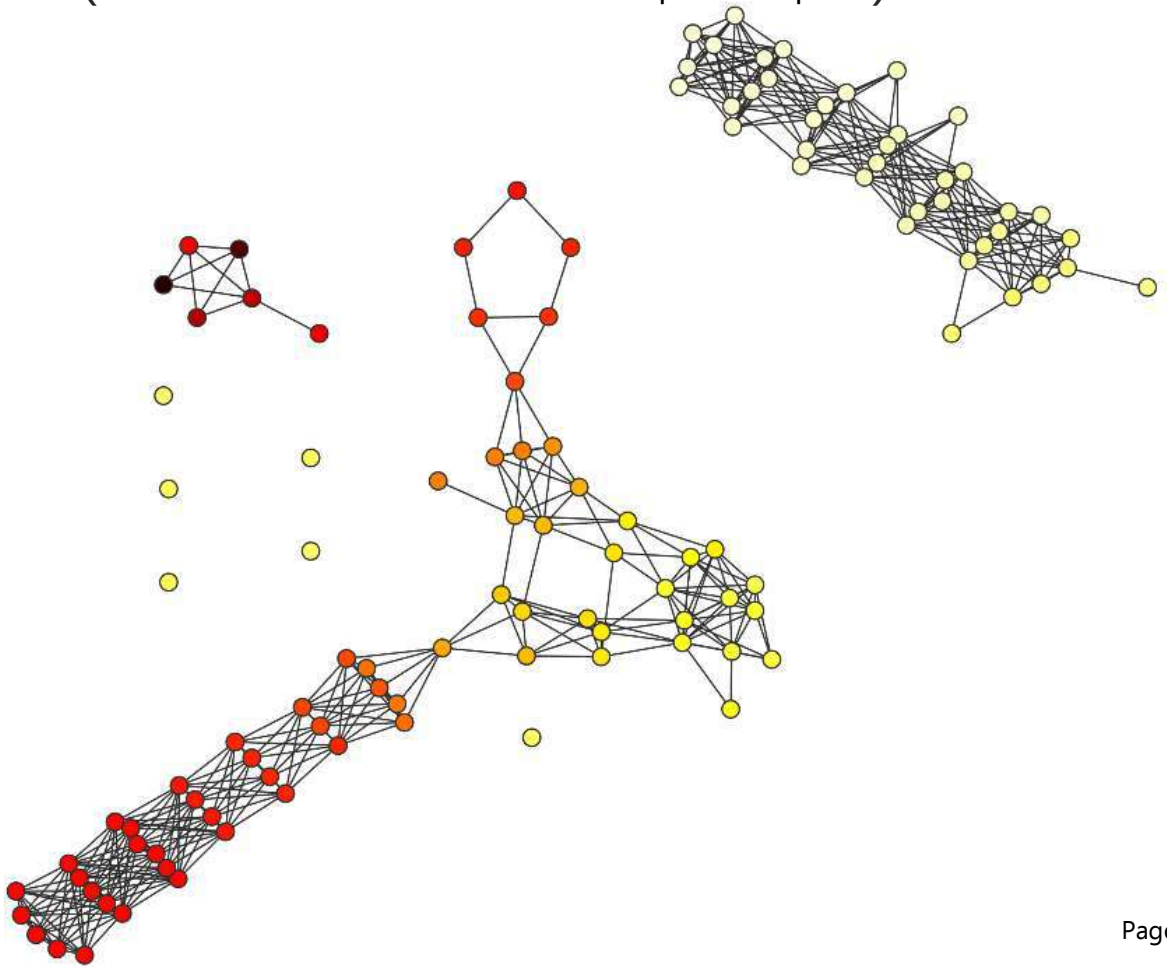
MαN graph

Both use projection function and covering

- Grounds on the Nerve Theorem
(global structure – Čech complex)



- Generate lots of metric graphs then select the most representative graph with robust geometry
(local structure – Vietoris–Rips complex)

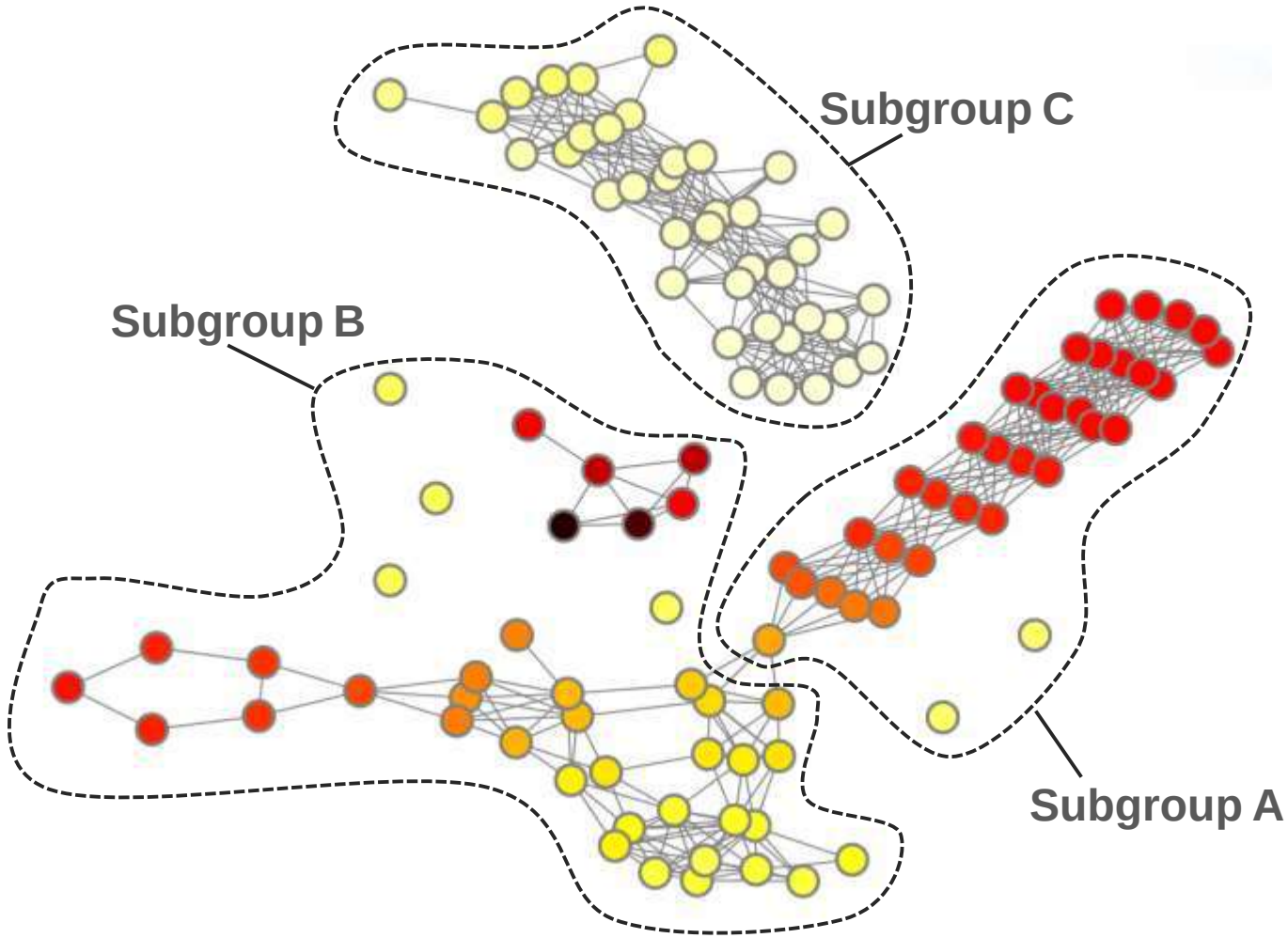


Community detection algorithms

Community detection

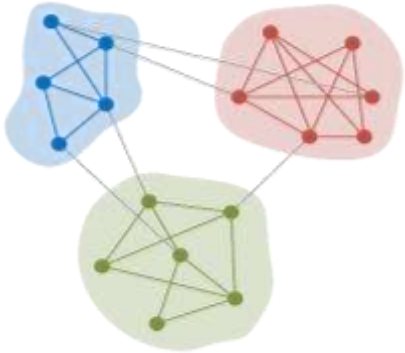
Community is a tightly-knit group of nodes with high density of within-group edges and a lower density of between-group edges

Nodes within communities are likely to **share common properties** and/or **play similar roles** in the graph



Community detection

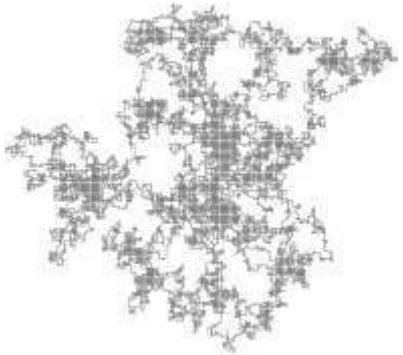
Different algorithms



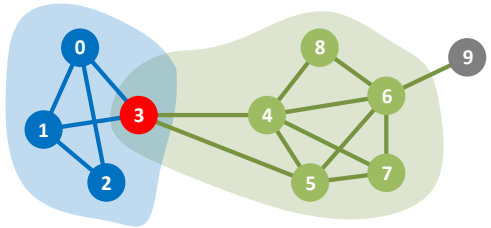
- Girvan-Newman algorithm

- Modularity maximization
  - ❖ Louvain method

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$



- Walk trap for random walks



- k-clique percolation

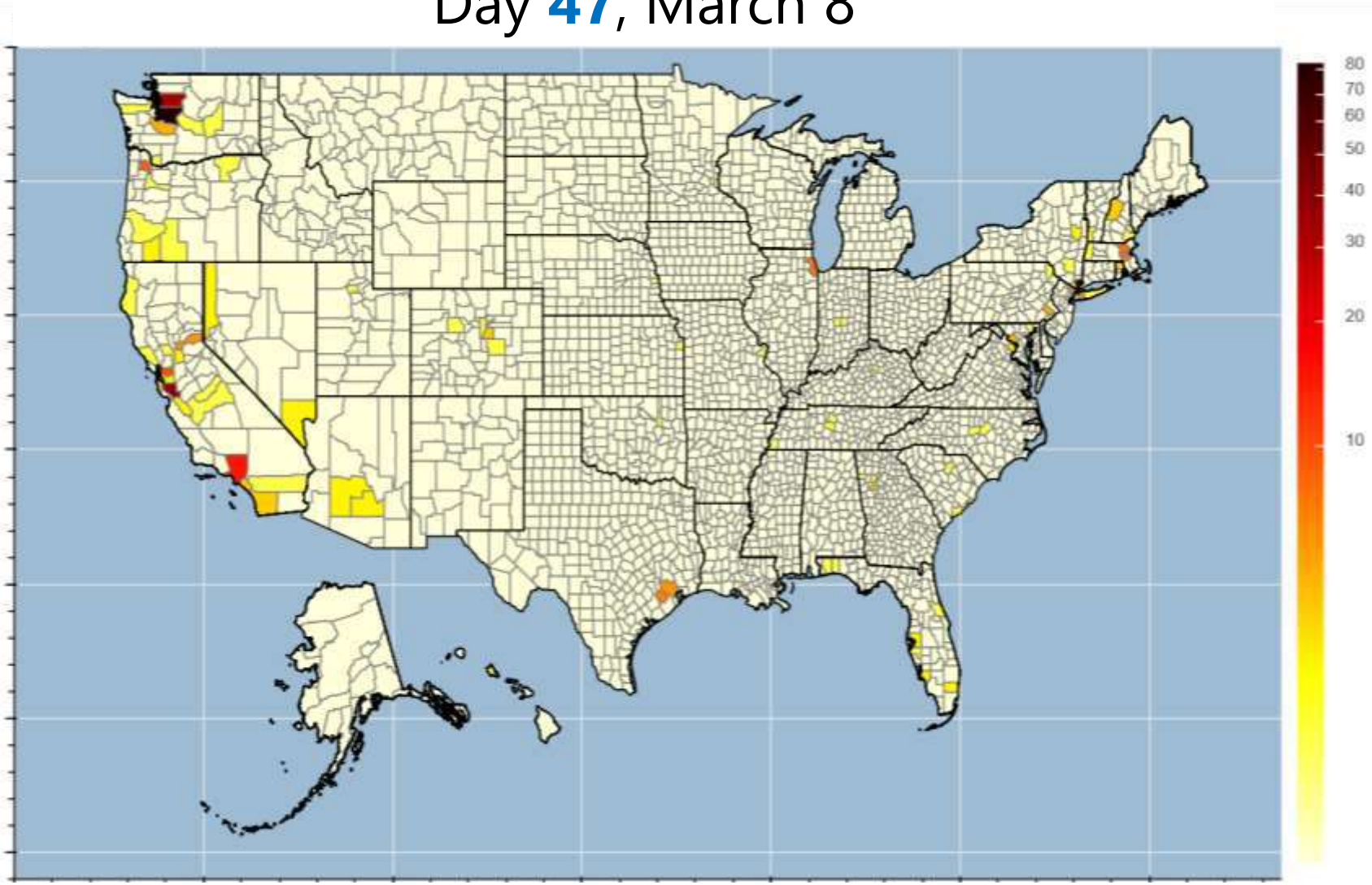
- …

UNDERSTANDING THE SPREAD OF COVID-19 IN THE UNITED STATES USING GRAPH-BASED DATA ANALYSIS

Understanding the spread of COVID-19 in the United States using topology and machine learning for time series, PHUSE EU-Connect-2020, on-line, November 9-13, 2020. https://www.lexjansen.com/phuse/2020/rw/PAP_RW03.pdf

Number of confirmed cases 451

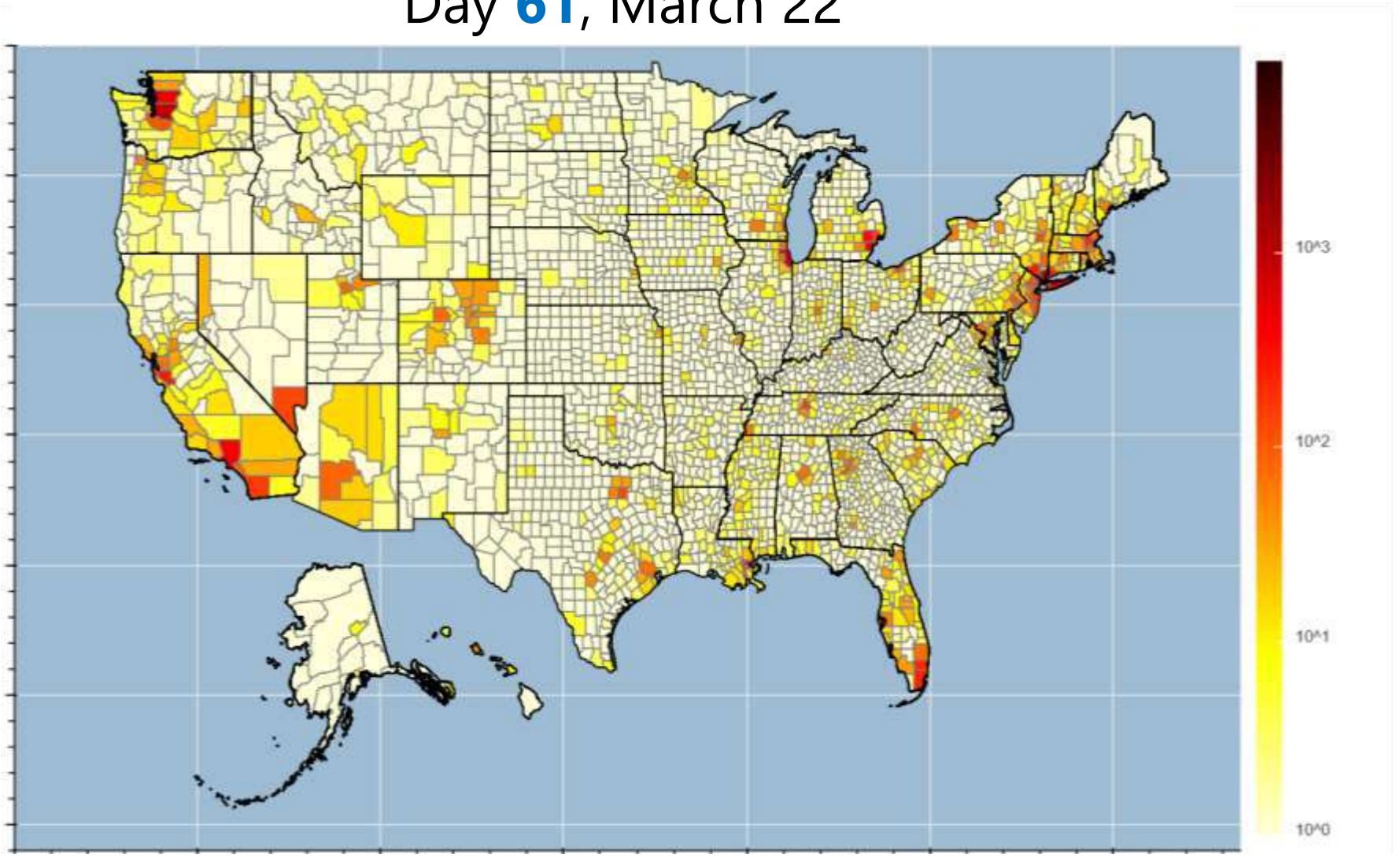
Day 47, March 8



Data source: https://github.com/CSSEGISandData/COVID-19

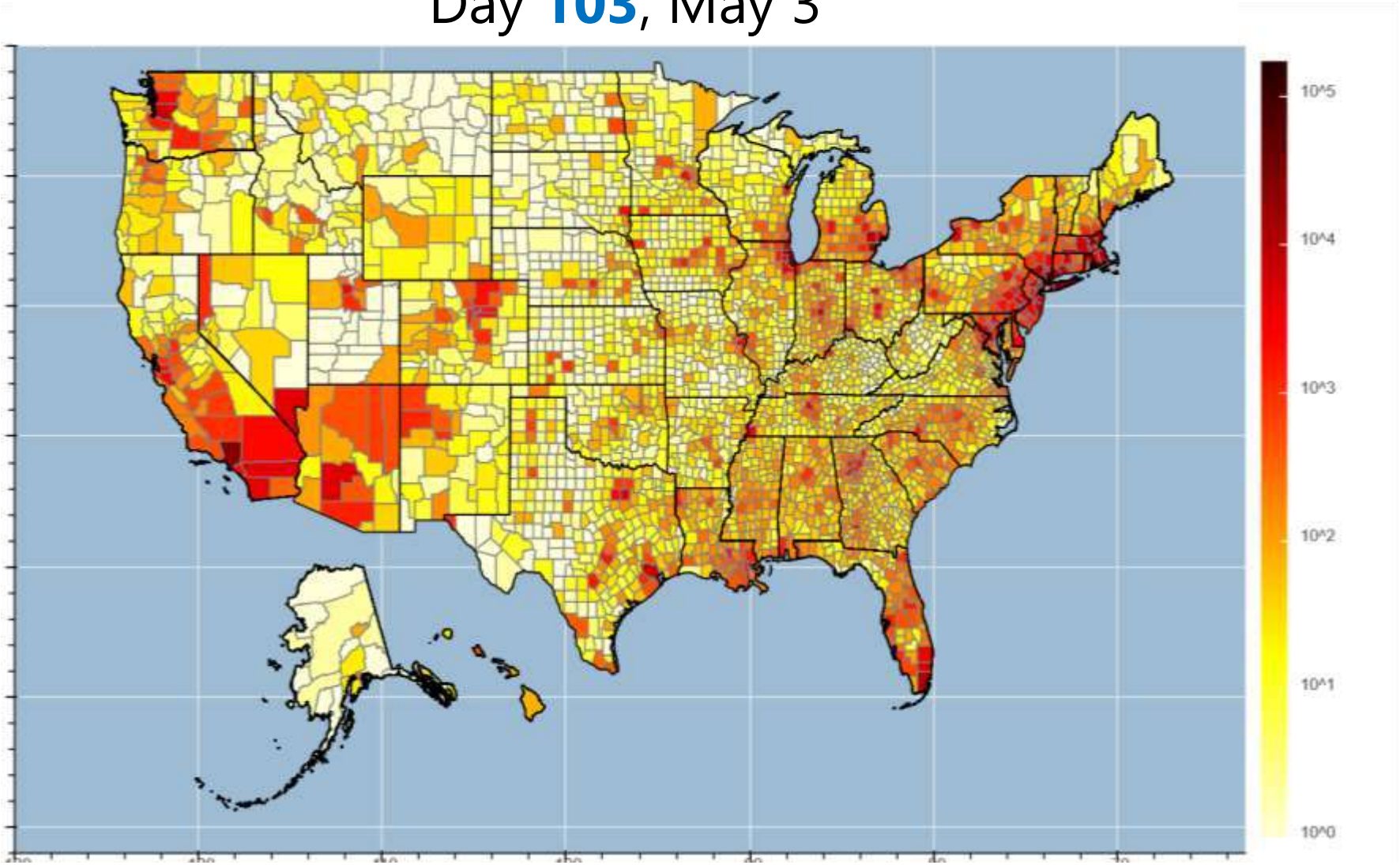
Number of confirmed cases 32,899

Day 61, March 22

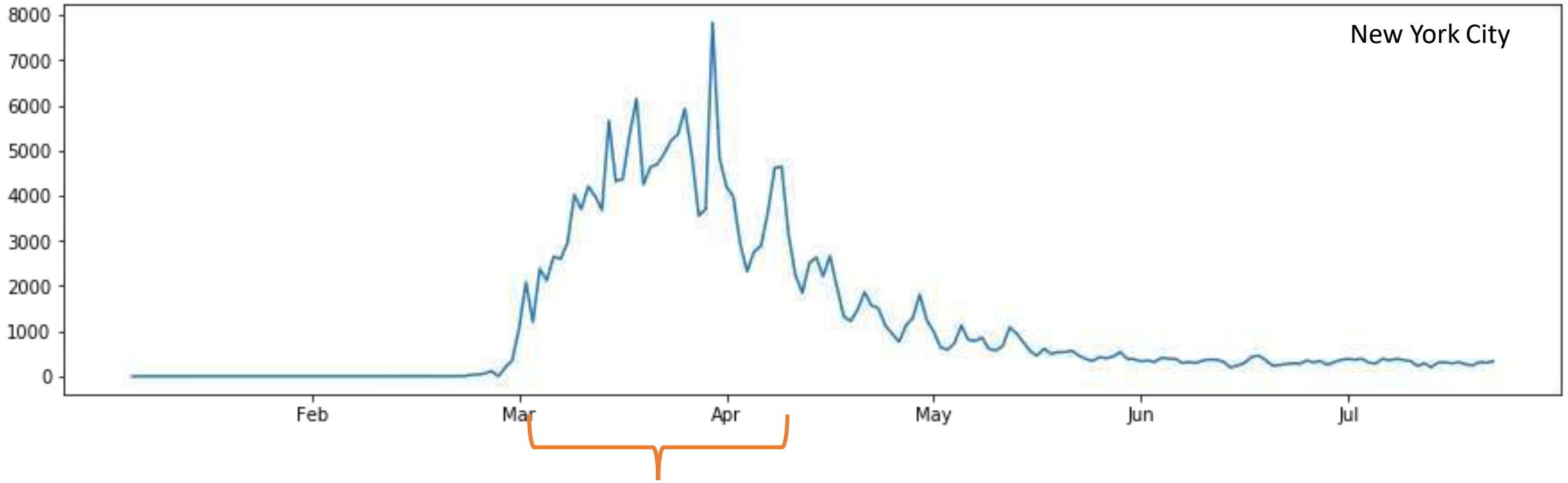


Number of confirmed cases 1,151,933

Day **103**, May 3



Time series observation interval



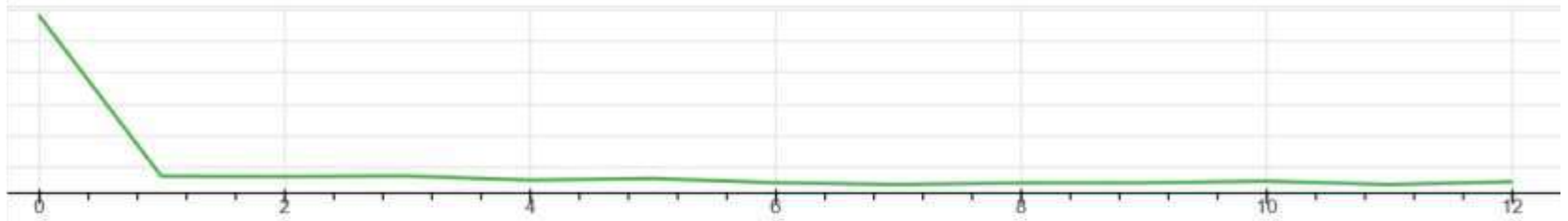
Interval=39 days

Start_day
2nd confirmed case
in the county

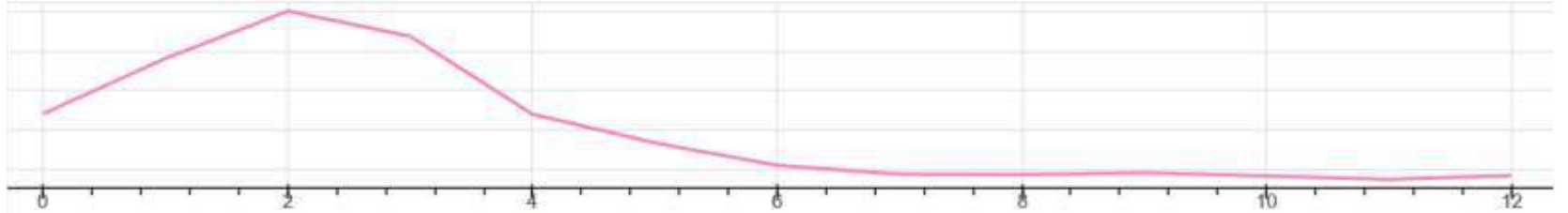
End_day
Release from
stay-at-home order

Epidemic curves of confirmed cases

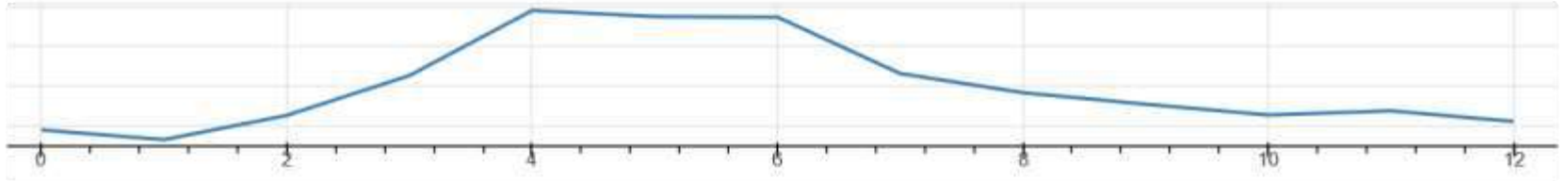
1



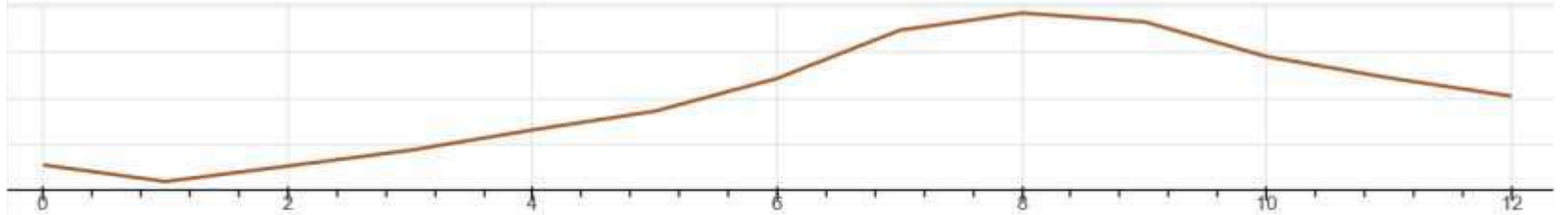
2



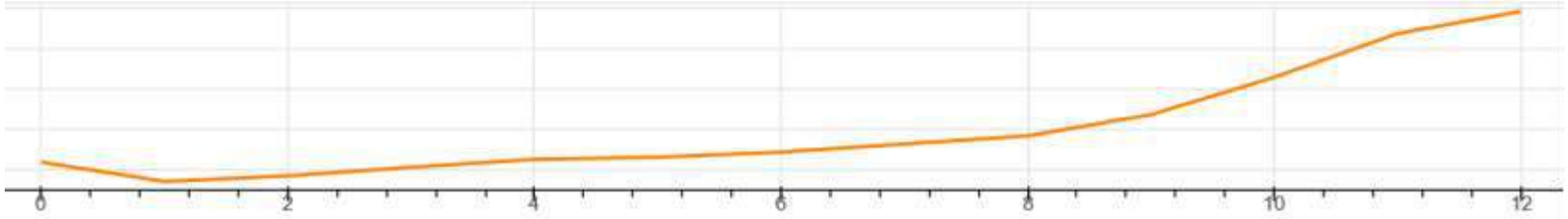
3



4



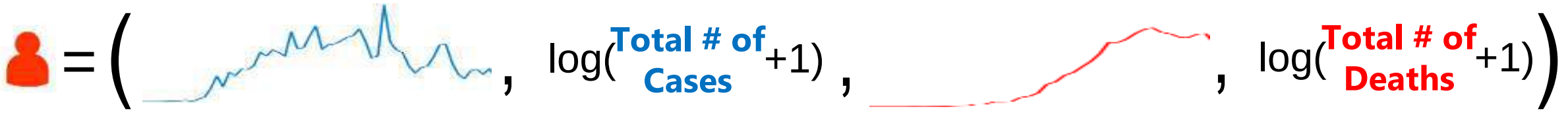
5



In our experiment:



- Each node corresponds to a county of the US
- Outcomes: all time series are within the selected observation interval of 39 days



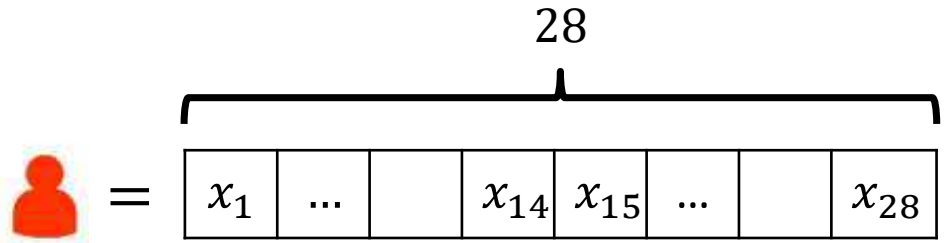
Confirmed cases, daily
(aggregated within sliding window of 3 days, normalized)

Death cases, daily
(aggregated within sliding window of 3 days, normalized)

Total number of entries per each county is 39/3 + 1 + 39/3 + 1 = **28**

Thus, the relevant information about the spread of COVID19 in a county is encoded by a **28-dimensional row vector**

(first 14 entries encode confirmed cases, second 14 entries encode deaths)



In our experiment:

- Stratifying bins arrange counties into several overlapping groups. Bins are chosen based on **density** and **centrality** of nodes with respect to outcomes.
- Similarity is then measured within each bin separately.



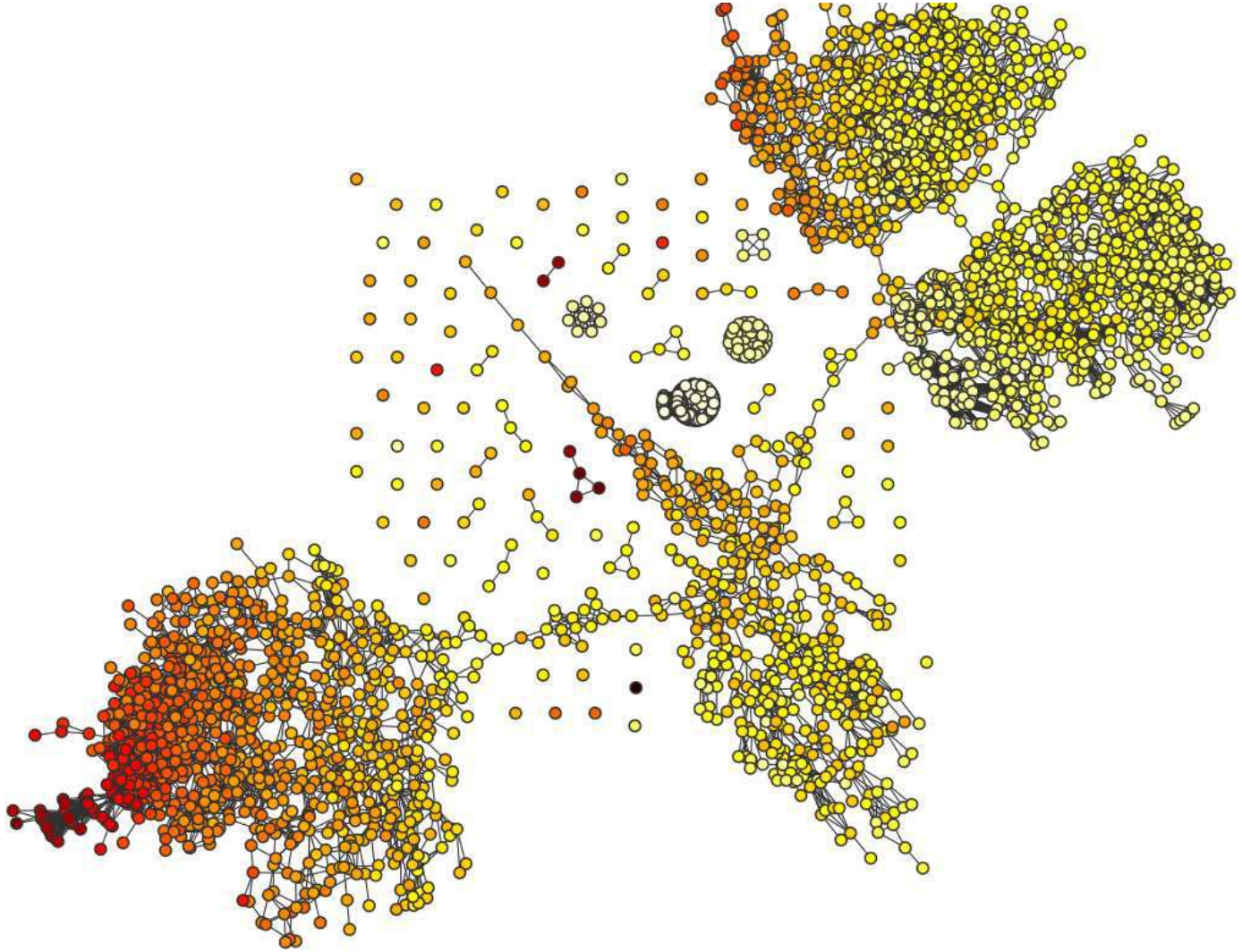
x_1	…		x_{14}	x_{15}	…		x_{28}

y_1	…		y_{14}	y_{15}	…		y_{28}

- **The function based on Euclidean distance** is used to measure the dissimilarity between counties

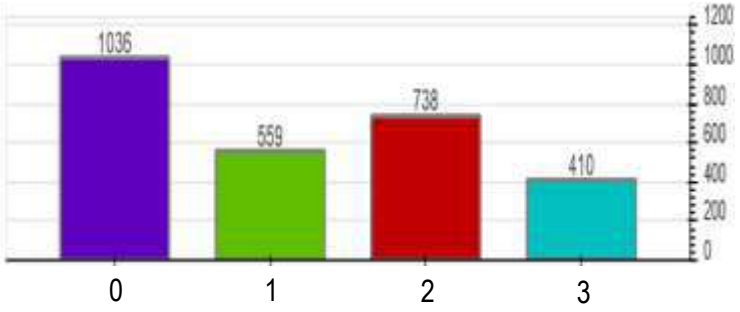
$$d(\text{●}, \text{●}) = \sqrt{(x_1 - y_1)^2 + \ldots + (x_{14} - y_{14})^2} + \sqrt{(x_{15} - y_{15})^2 + \ldots + (x_{28} - y_{28})^2}$$

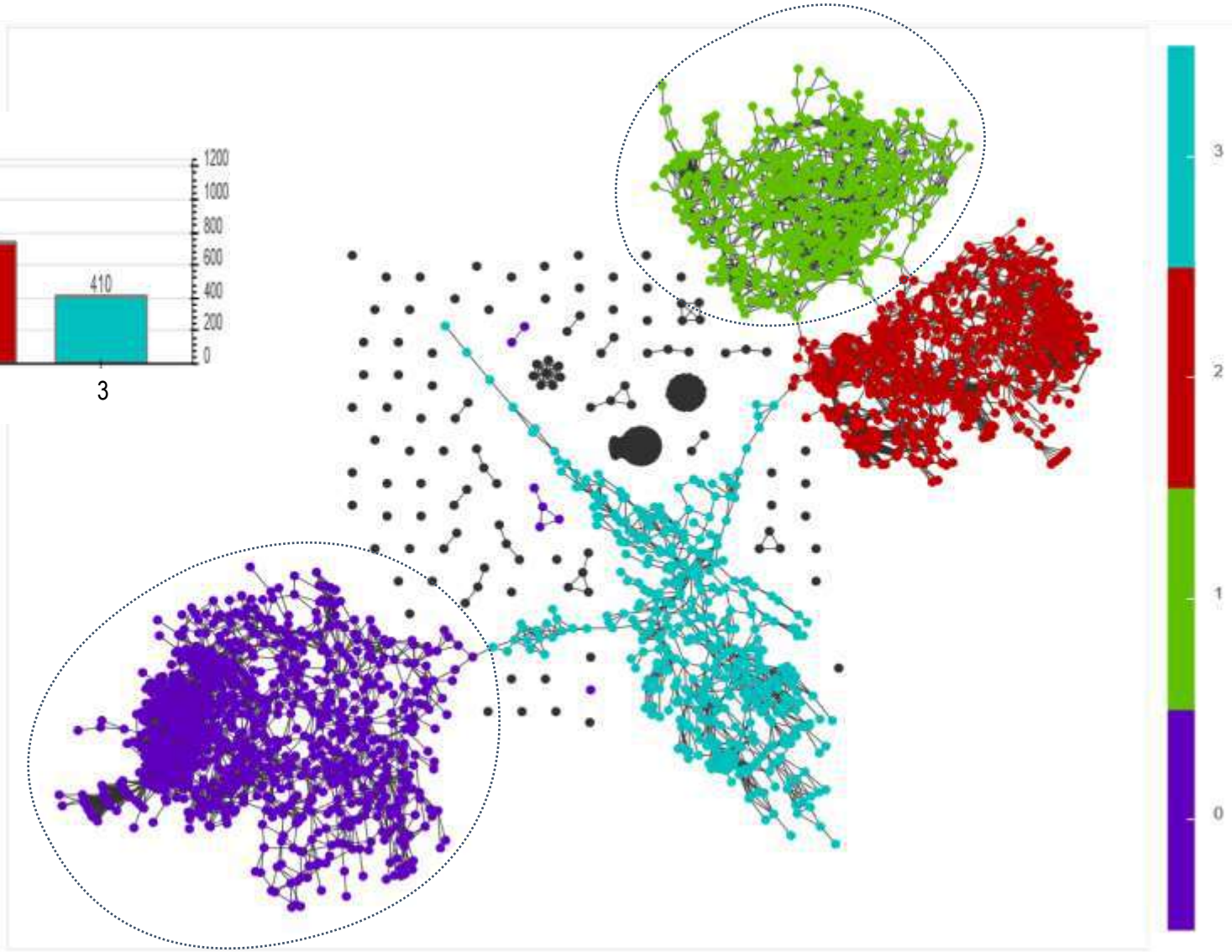
Graph-based model of COVID-19 spread over the US



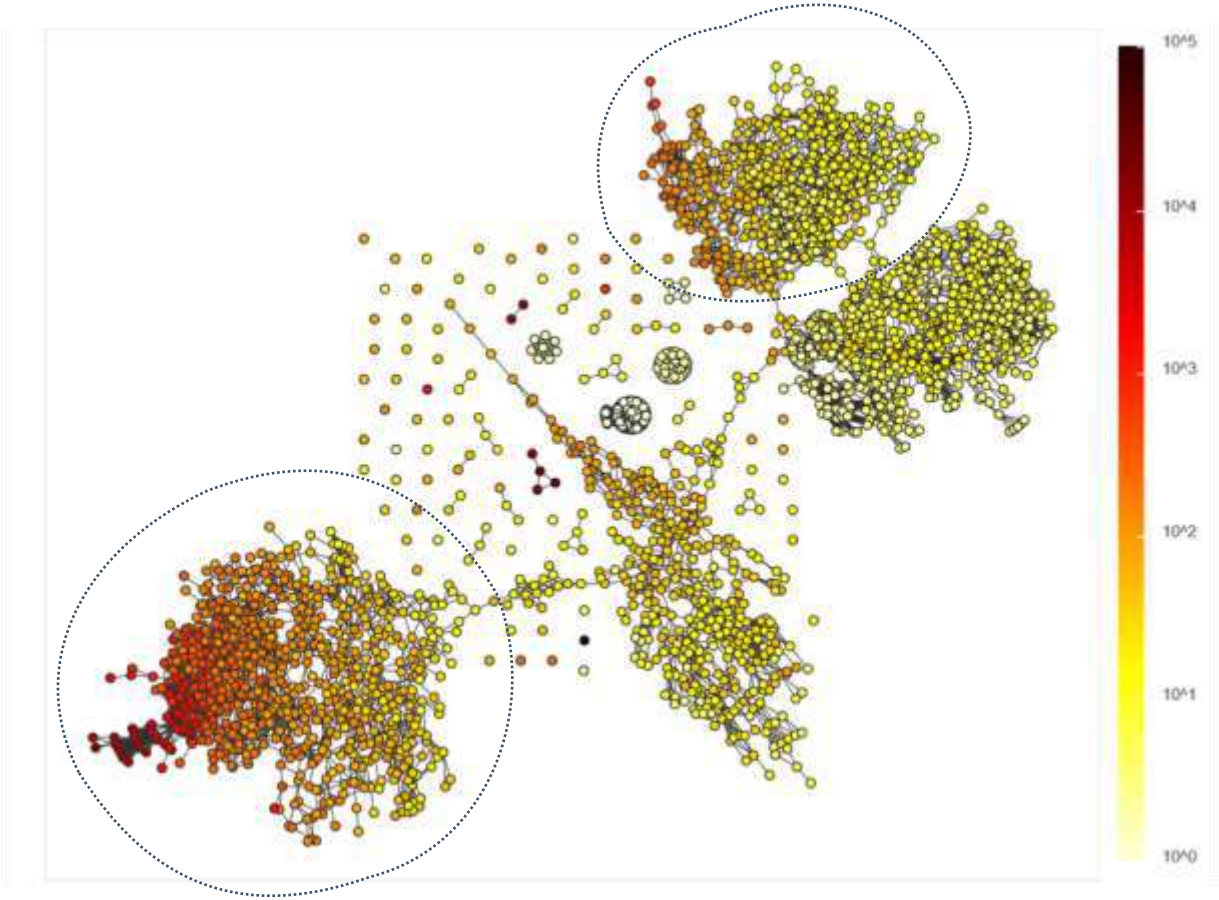
- Each node on the **graph** corresponds to one county of the United States (total **3,142 nodes**)
- Two nodes are connected with the edge if they are **similar** in terms of predefined outcomes
- Outcomes integrate the number of confirmed **cases and deaths** within a **39-day observation interval** from the start of the pandemic
- **Similarity** is measured by a modified Euclidean distance function between time series of outcomes

Major communities

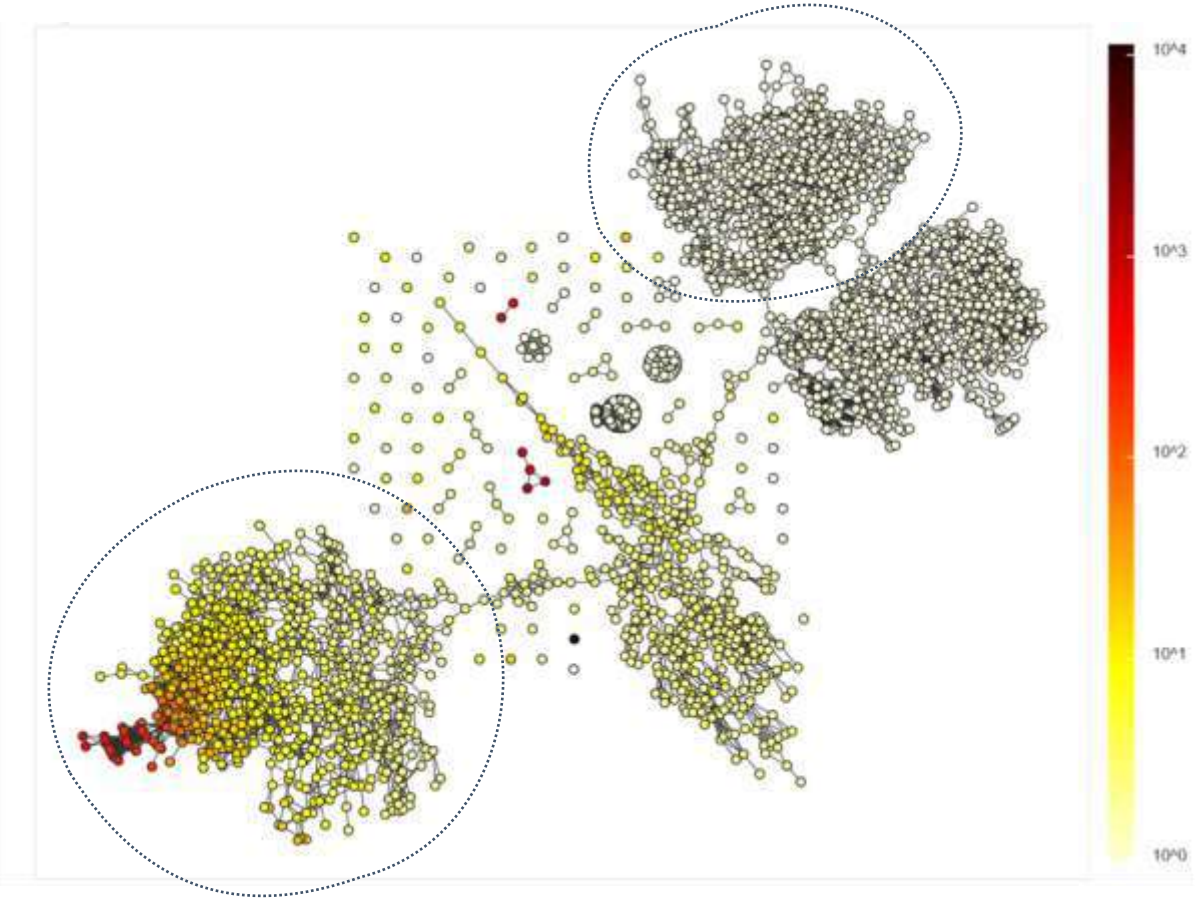




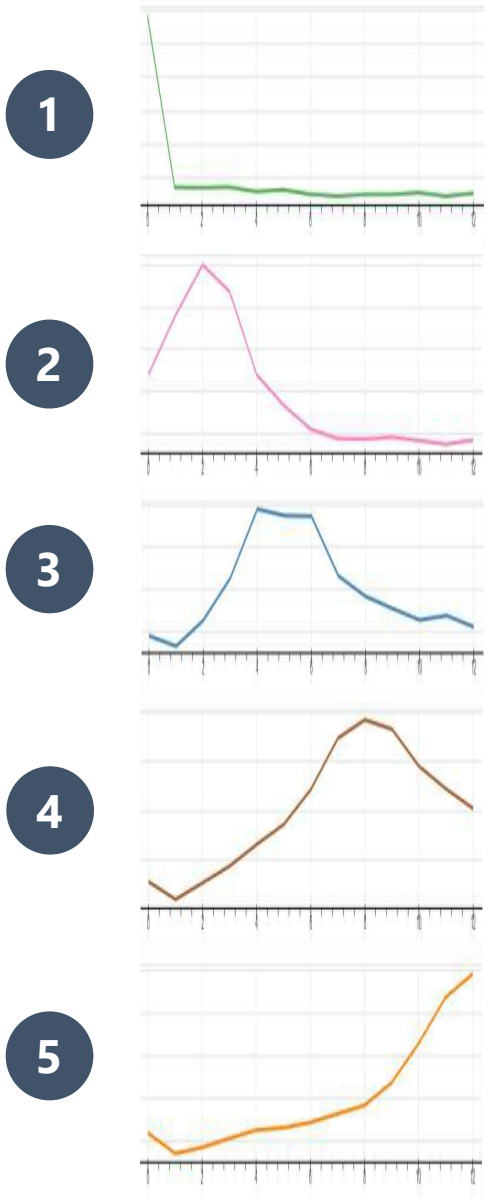
Number of confirmed cases

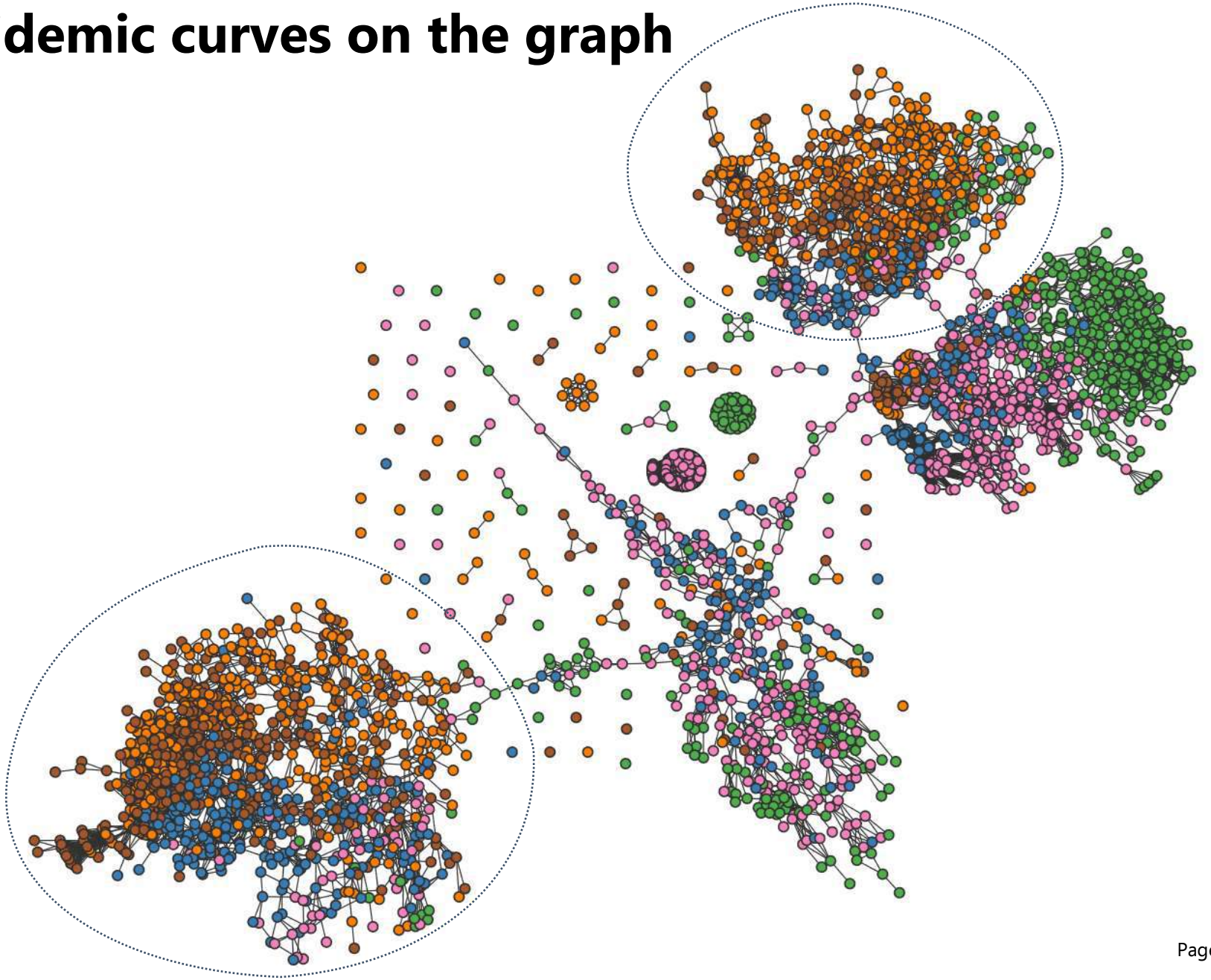


Number of deaths



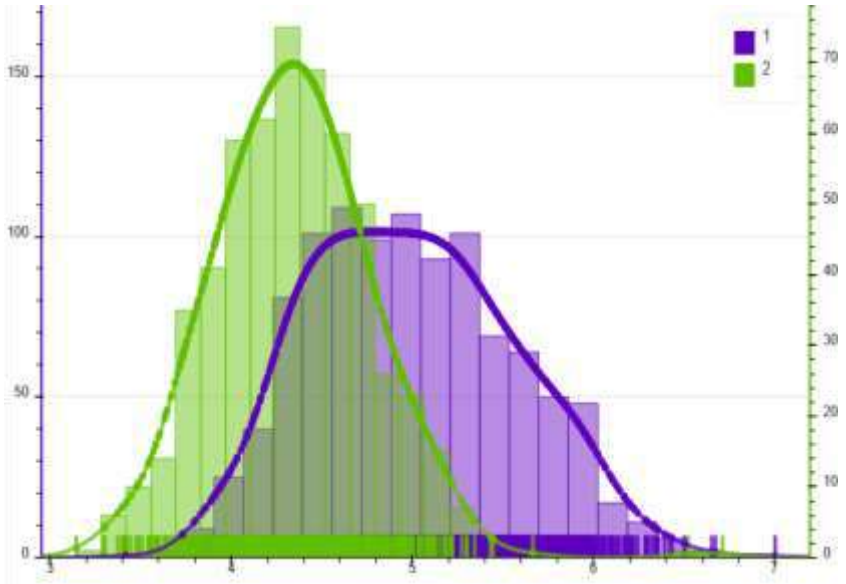
Epidemic curves on the graph





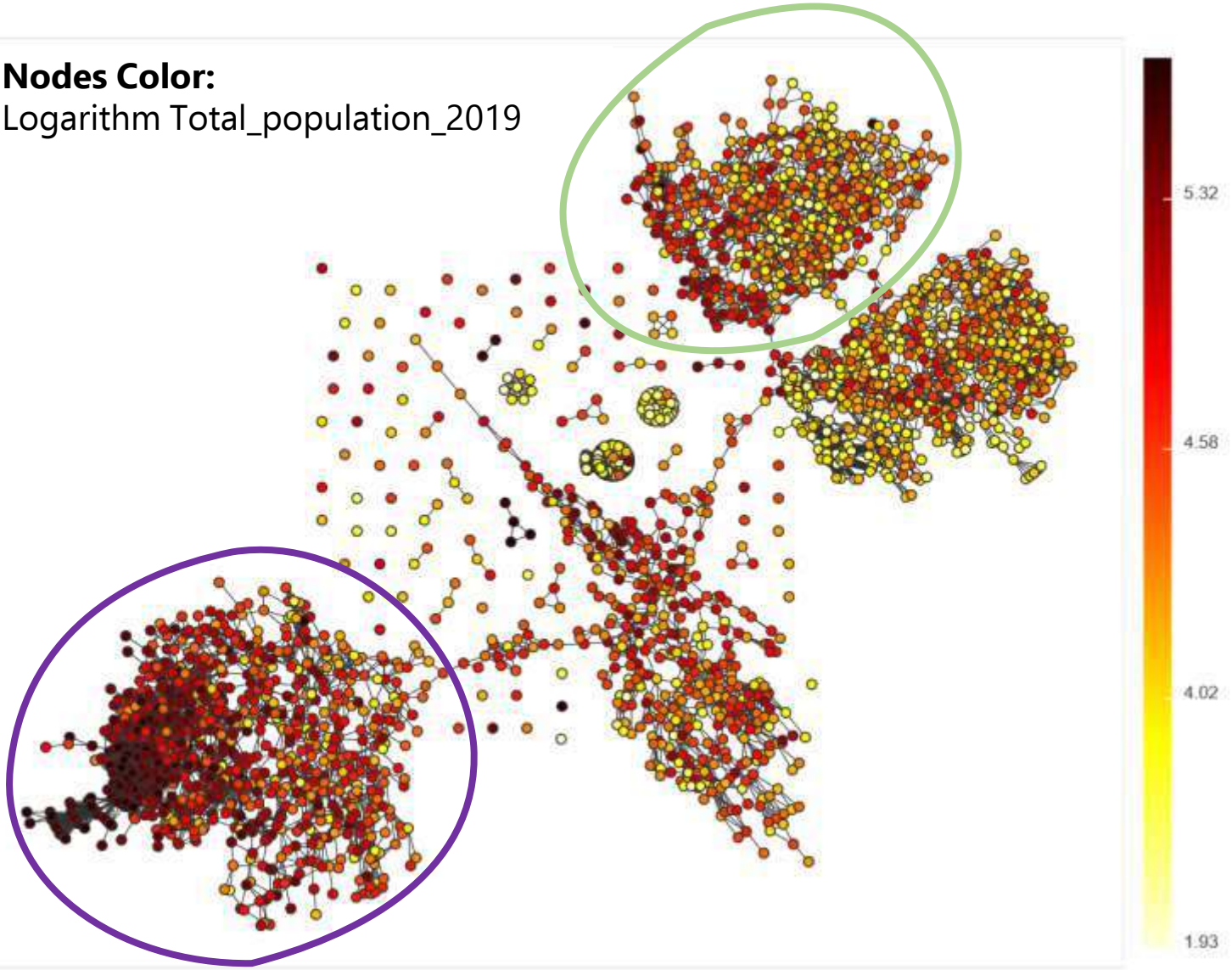
Population

P-Value 1.4e-97



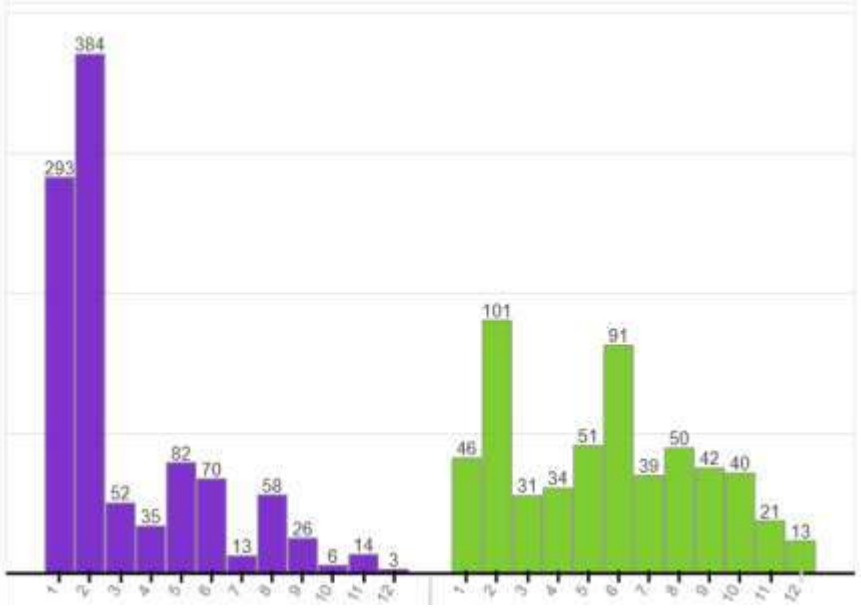
Count	Mean	Median	Min	Max	Std
1036	4.999	4.974	3.416	7.002	0.572
559	4.343	4.323	3.146	6.712	0.439

Nodes Color:
Logarithm Total_population_2019



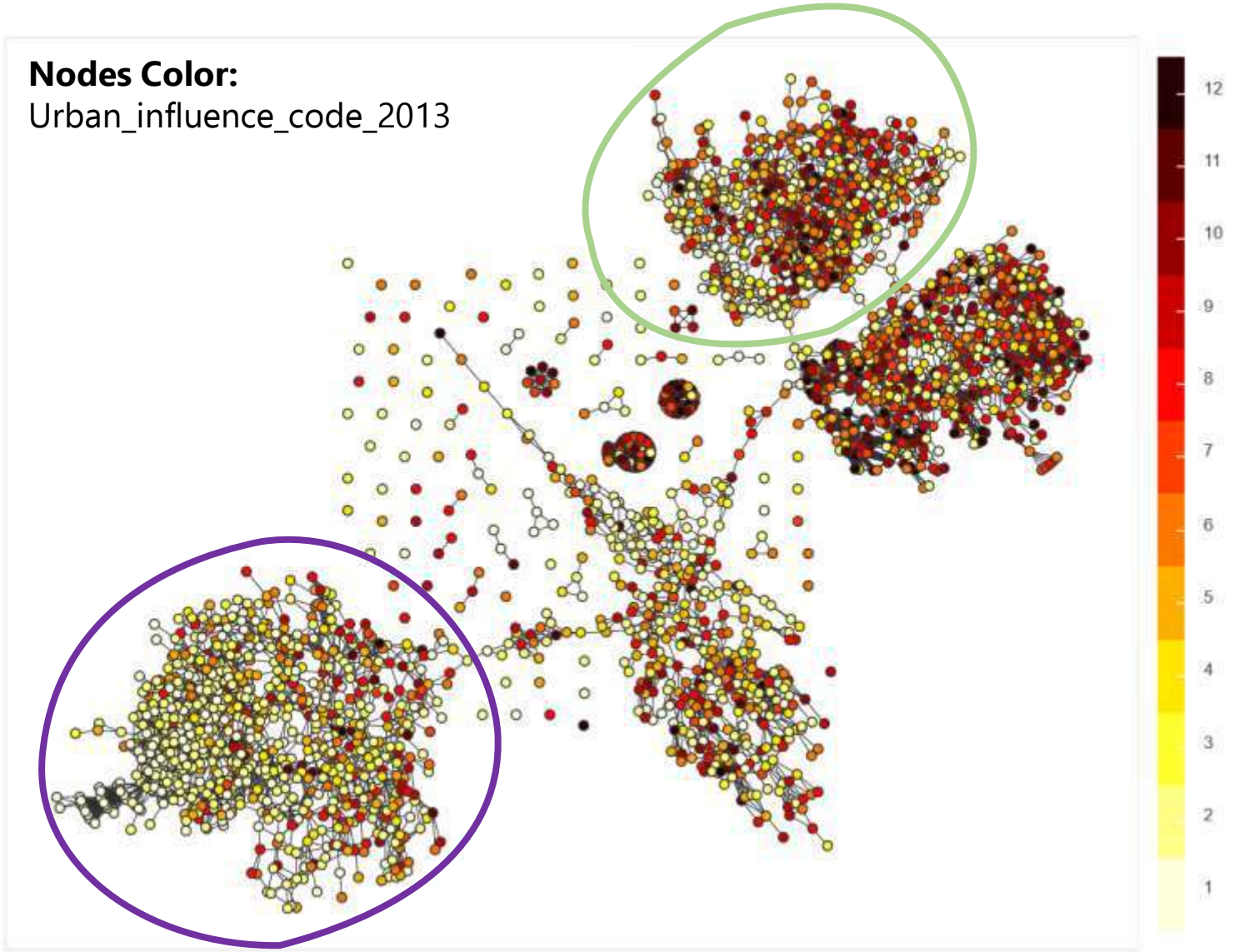
Urban Influence Code

P-Value 1.5e-56



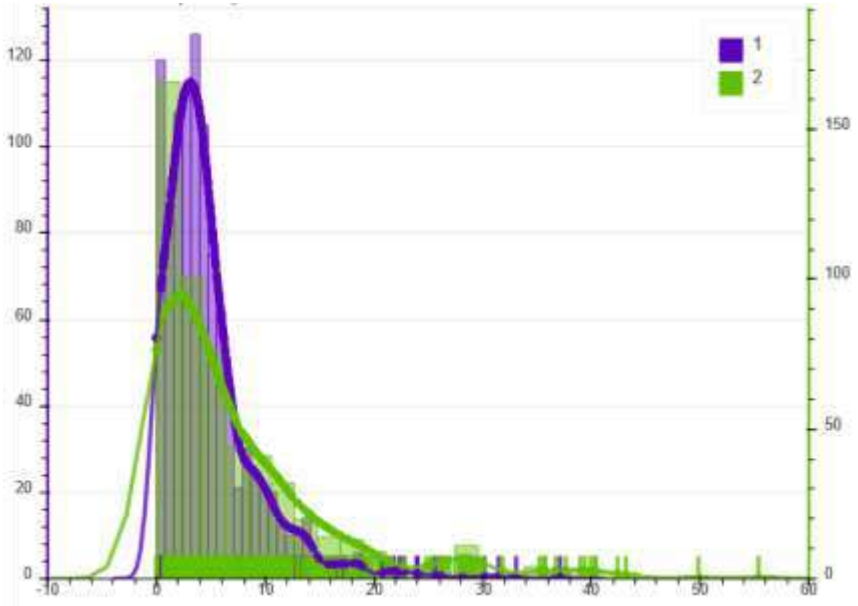
1	2	3	4	5	6	7	8	9	10	11	12
293	384	52	35	82	70	13	58	26	6	14	3
46	101	31	34	51	91	39	50	42	40	21	13

Nodes Color:
Urban_influence_code_2013



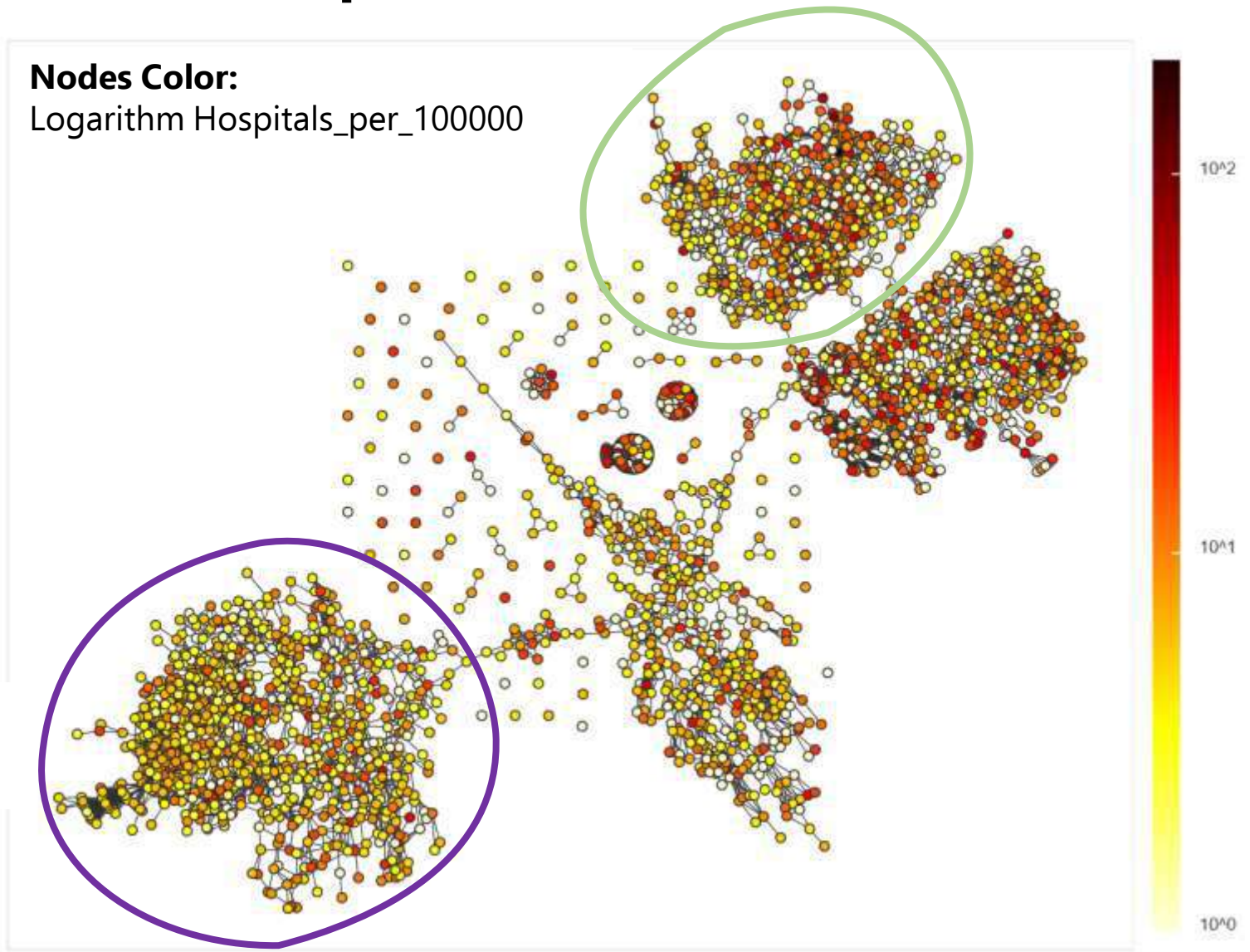
Number of hospitals

P-Value 0.002



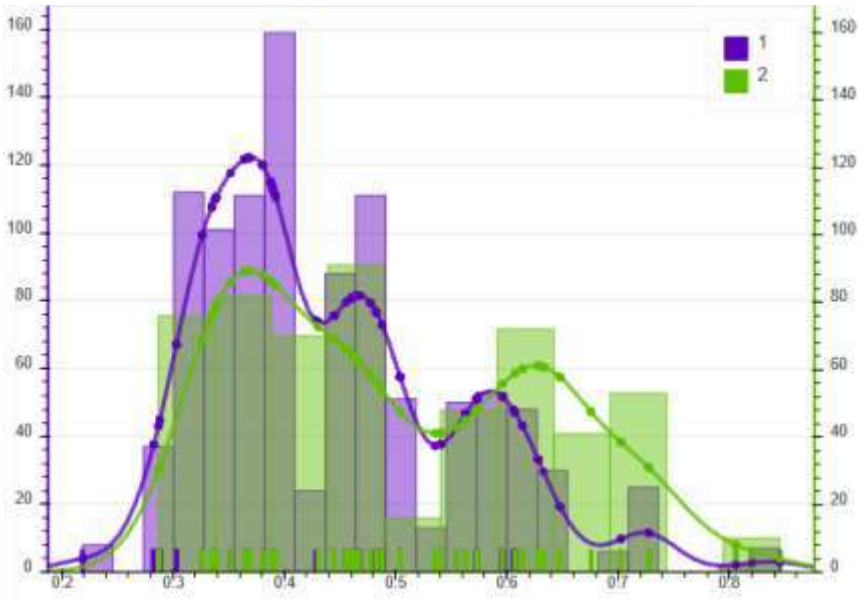
Count	Mean	Median	Min	Max	Std
1036	4.847	3.762	0	37.116	4.524
559	7.9	4.569	0	200.602	12.563

Nodes Color:
Logarithm Hospitals_per_100000



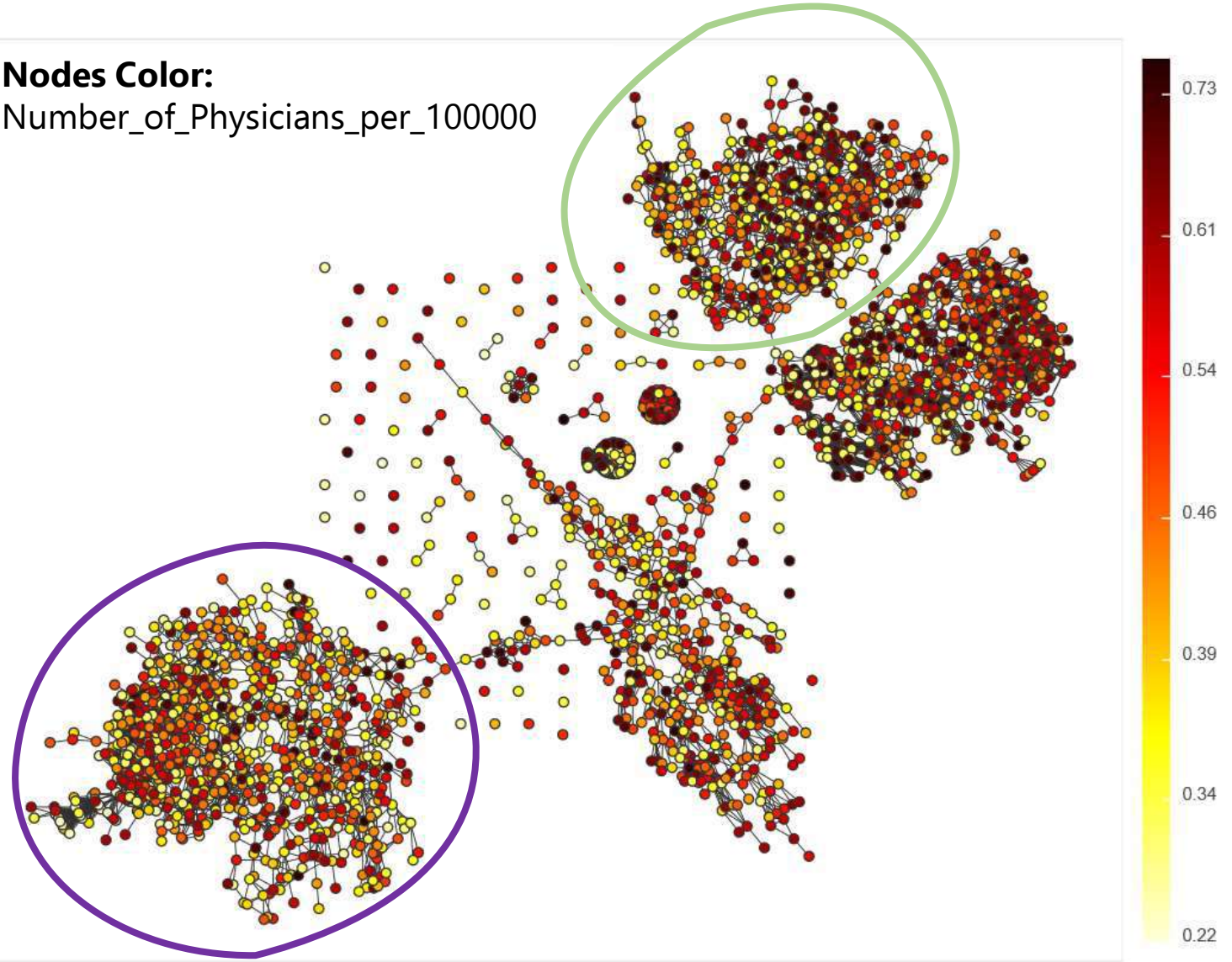
Number of General Practice Physicians

P-Value 1.1e-14



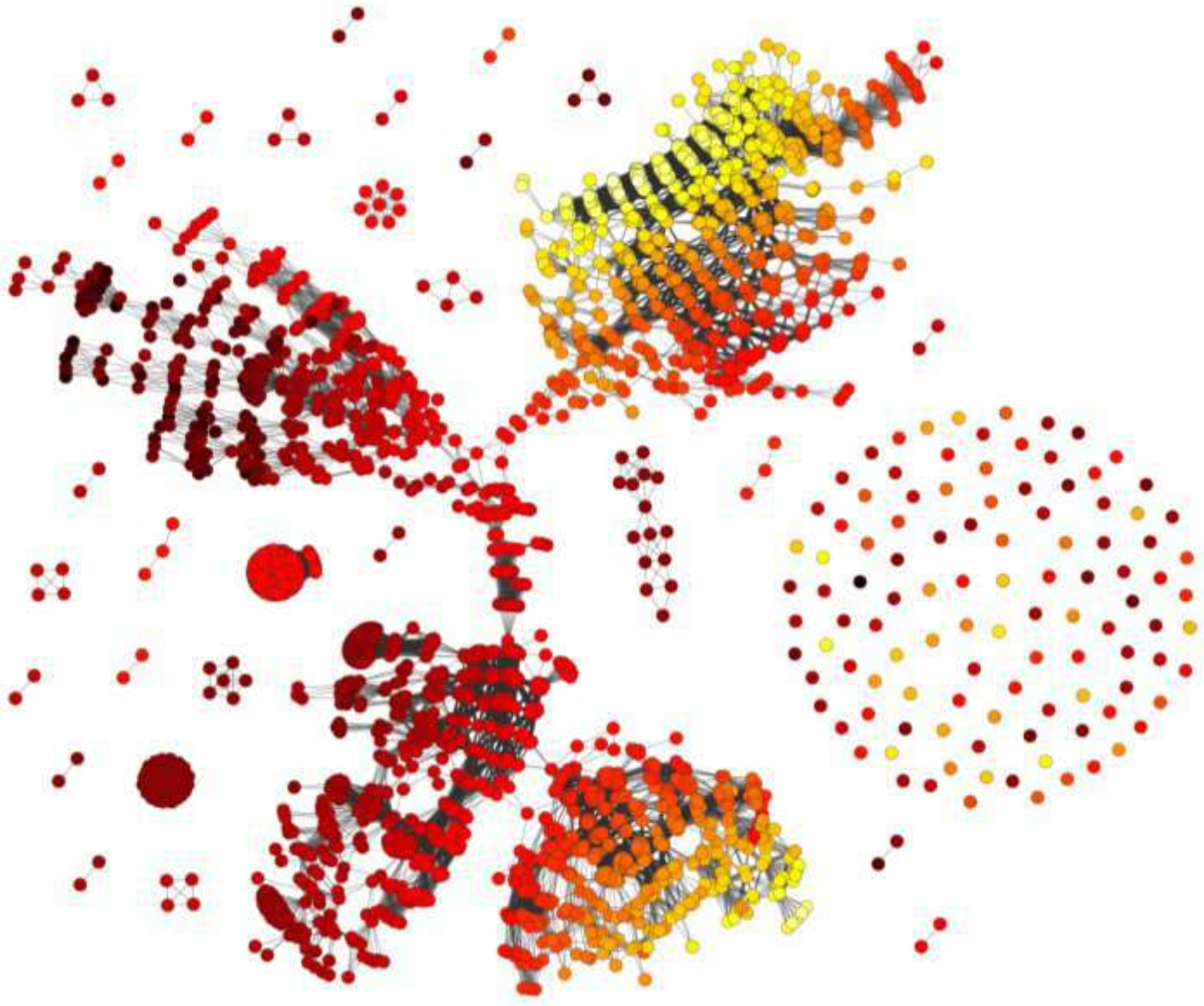
Count	Mean	Median	Min	Max	Std
1036	0.443	0.392	0.219	0.845	0.115
559	0.498	0.468	0.287	0.845	0.136

Nodes Color:
Number_of_Physicians_per_100000



The COVID-19 MαN graph with spectral α-scale selection

MαN graph with spectral α-scale selection of COVID-19 spread over the US



The min matrices used for filtration and for establishing the optimal scales are defined by a functional **|C|·|E|**
(**|C|** - number of components, **|E|** number of edges)

The graph has also

- ✓ High clustering score
  (random walk mixing rate = 0.99997)
- ✓ Visible 4 communities